OPTIMAL ESTIMATES FOR DIRECT IMAGES: PRODUCT DESCENT AND TOP-DEGREE RIGIDITY

ZHIJIAN CHEN, ZHUO LIU, AND CHENGHAO QING

ABSTRACT. We prove product descent of optimal L^2 estimates for prescribed direct-image metrics. For arbitrary singular semipositive line metrics, a rigidity theorem gives non-product adjoint descent in top fiber degree $q = n$. We identify the smooth adjoint metric with the Mourougane–Takayama metric and give counterexamples to unrestricted descent.

CONTENTS

1. Introduction 1
2. Curvature estimates and weighted cohomology 4
3. Smooth product descent 9
4. Top direct images and the smooth adjoint case 13
5. Singular product metrics 17
6. Top adjoint descent for arbitrary singular semipositive metrics 22
7. Counterexamples to unrestricted descent 28
8. Scalar-factor metrics and minimal extension 32
References 34

1. INTRODUCTION

Let $\pi : X \to Y$ be a proper holomorphic submersion between complex manifolds, with compact fibers X_y of dimension n and base dimension m. Fix Kähler metrics and a Hermitian holomorphic line bundle (L, h) on X. For smooth h, the fiber cohomology $H^q(X_y, \Omega^p_{X_y} \otimes L_y)$ carries its harmonic norm. For a metric with plurisubharmonic weights, the relevant sheaf is

$$\mathcal{F}_h^{p,q} = R^q\pi_*(\Omega^p_{X/Y} \otimes L \otimes \mathcal{I}(h)),$$

and the prescribed norm is the minimum weighted L^2 norm of a Dolbeault representative, wherever the fiberwise identification is defined. The multiplier ideal and the choice of norm are part of the statement.

The positivity of direct images developed from Hodge-theoretic semipositivity, the theorems of Fujita and Kawamata [Fuj78, Kaw81], Viehweg's weak positivity [Vie83], and Kollár's higher-direct-image theorems [Kol86a, Kol86b]. Takegoshi treated Nakano semipositive coefficients under proper Kähler morphisms using harmonic methods [Take95]. Mourougane–Takayama studied Hodge metrics on adjoint direct images [MT07]; Berndtsson proved Nakano semipositivity of the natural L^2 metric on $f_*(K_{X/Y} \otimes L)$ for a smooth semipositive line metric [Ber09]. Mourougane–Takayama obtained the higher-direct-image version [MT08] and extended the induced tautological metrics across singular fibers [MT09]. Curvature formulas for other Hodge degrees were

Date: 30 October 2026.
2020 *Mathematics Subject Classification.* Primary 32L10; Secondary 32G05, 32L20, 32W05.
Key words and phrases. Higher direct images, optimal L^2 estimates, singular Hermitian metrics, multiplier ideals, harmonic metrics, relative duality.

subsequently developed by, among others, Liu–Yang, Naumann, Berndtsson–Păun–Wang, and Choi–Schumacher [LY14, Nau21, BPW22, CS24].

For singular coefficients, Berndtsson–Păun established positivity of relative Bergman metrics [BP08], and Păun–Takayama developed singular metric positivity for twisted pluricanonical direct images [PT18]. These methods enter Cao–Păun's work on algebraic fiber spaces over abelian varieties [CP17]; Hacon–Popa–Schnell give a detailed treatment of the adjoint direct-image metric and its minimal extension property [HPS18]. The formulation of positivity by optimal $\bar\partial$ estimates was developed in [DNWZ23, Wat25b]. For ordinary adjoint direct images, singular Nakano positivity is proved in [Wat25a, IMW24]; the latter allows arbitrary singular semipositive line metrics and proper Kähler fibrations. Takayama proves singular Griffiths semipositivity and the minimal extension property for higher direct images with smooth Nakano semipositive coefficients [Tak24].

The multiplier ideal in a singular higher direct image is essential. Demailly–Peternell–Schneider prove a hard Lefschetz theorem with multiplier ideals [DPS01], and Wu studies the resulting parallel sections [Wu21]. Matsumura establishes relative injectivity and torsion-freeness for arbitrary singular semipositive metrics [Mat22]. Our top-degree argument uses instead the rigidity forced by nonvanishing, together with relative duality and the divisorial decomposition of a positive current [RRV71, Siu74].

The product descent question compares an optimal estimate in total degree $(a+p,b+q)$ with the estimate in base degree (a,b) for the prescribed direct-image metric. The smooth and weighted optimal estimates are defined in Definitions 2.1 and 2.4; $\mathscr{E}(H)$ denotes the sheaf of locally square-integrable holomorphic sections. We use the ranges

$$\begin{array}{lllll} \text{(i)} & a=m, & p=n, & 1\le b\le m, & 0\le q\le n,\\ \text{(ii)} & b=m, & q=n, & 1\le a\le m, & 0\le p\le n. \end{array} \tag{1.1}$$

Theorem 1.1 (Smooth product descent)**.** *Let (Y,ω_Y) be complete Kähler of dimension $m\ge 1$, admitting a positive Hermitian holomorphic line bundle. Let (Z,ω_Z) be smooth projective of dimension n, and set*

$$X=Y\times Z,\quad \pi=\mathrm{pr}_Y,\quad \omega_X=\omega_Y+\omega_Z,\quad (L,h)=(G,h_G)\boxtimes(M,h_M),$$

where G and M are holomorphic line bundles on Y and Z, respectively, and both metrics are smooth. In either range (1.1), if (L,h) satisfies the optimal $(a+p,b+q)$ estimate, then

$$(\mathcal{F},H)=(R^q\pi_*(\Omega^p_{X/Y}\otimes L),H)$$

satisfies the optimal (a,b) estimate for its prescribed harmonic metric.

In range (ii), Proposition 3.2 allows an arbitrary smooth metric on $G\boxtimes M$, with the same product geometry. Singular semipositive product metrics give the following result.

Theorem 1.2 (Singular product descent)**.** *Let (Y,ω_Y) be complete Kähler of dimension $m\ge 1$ and (Z,ω_Z) compact Kähler of dimension n. On $X=Y\times Z$, with $\pi=\mathrm{pr}_Y$, let*

$$(L,h)=(G,h_G)\boxtimes(M,h_M),$$

where G and M are holomorphic line bundles on Y and Z with arbitrary psh metric weights. Fix $0\le p,q\le n$ and define

$$V=H^q(Z,\Omega^p_Z\otimes M\otimes\mathcal{I}(h_M)),\quad E_0=G\otimes V,\quad H=h_G\otimes H_V.$$

Here H_V is the minimum weighted L^2 norm of Dolbeault representatives, whose existence is proved in Lemma 2.6. There is a canonical identification

$$\begin{aligned}\mathcal{F}&:=R^q\pi_*(\Omega^p_{X/Y}\otimes L\otimes\mathcal{I}(h))\\ &=(G\otimes\mathcal{I}(h_G))\otimes V=\mathscr{E}(H).\end{aligned} \tag{1.2}$$

For almost every y, H_y is the prescribed weighted fiberwise quotient norm. The pair in (1.2) satisfies the weighted optimal estimates in all degrees

$$(m,b), \quad 1 \le b \le m, \qquad \text{and} \qquad (a,m), \quad 1 \le a \le m,$$

for every p, q. The metric H on the torsion-free sheaf $\mathcal{F}$ is Griffiths semipositive and has the minimal extension property.

The top adjoint degree admits a non-product result for completely arbitrary singular semipositive coefficient metrics.

Theorem 1.3 (Rigidity and descent in top adjoint degree)**.** *Let $\pi : X \to Y$ be a proper surjective holomorphic submersion between connected complex manifolds, with connected fibers of dimension $n \ge 1$. Fix a Kähler form ω_X on X, put $\omega_y = \omega_X|_{X_y}$, and let (L,h) be a holomorphic line bundle with arbitrary psh metric weights. Set*

$$\mathcal{F}_n = R^n\pi_*(K_{X/Y} \otimes L \otimes \mathcal{I}(h)).$$

Either $\mathcal{F}_n = 0$, or there exist an effective divisor D on X and a holomorphic line bundle G on Y with a singular Hermitian metric h_G having psh local weights such that

$$(L,h) \simeq (\mathcal{O}_X(D), h_D) \otimes \pi^*(G, h_G). \tag{1.3}$$

Here s_D is the canonical section of $\mathcal{O}_X(D)$ and h_D is the divisorial metric characterized by $|s_D|_{h_D} = 1$ off D. In the second case,

$$\mathcal{I}(h) = \mathcal{O}_X(-D) \cdot \pi^{-1}\mathcal{I}(h_G)\mathcal{O}_X, \qquad \mathcal{F}_n \simeq G \otimes \mathcal{I}(h_G). \tag{1.4}$$

The prescribed weighted fiberwise quotient norm is, almost everywhere,

$$H = V^{-1}h_G, \qquad V = \int_{X_y} \frac{\omega_y^n}{n!} > 0, \tag{1.5}$$

where V is independent of y, and the trace of a top Dolbeault class is normalized by integration. In particular, $\mathcal{F}_n = \mathscr{E}(H)$ as a subsheaf of G.

If (Y, ω_Y) is complete Kähler of dimension $m \ge 1$, then $(\mathcal{F}_n, H)$ satisfies the singular optimal (m,b) estimate for every $1 \le b \le m$.

For smooth vector coefficients, isometric Serre duality gives the following complementary statement.

Theorem 1.4 (Top direct images under infinitesimal triviality)**.** *Let $\pi : X \to Y$ be a proper surjective holomorphic submersion of relative dimension n between Kähler manifolds. Assume that all fiberwise Kodaira–Spencer maps vanish. Let (E,h) be a smooth Hermitian holomorphic vector bundle such that (E^*, h^*) is Nakano seminegative, and assume that $h^n(X_y, E_y)$ is locally constant. Then the prescribed harmonic metric on $R^n\pi_*E$ is smooth and dual Nakano semipositive, that is, its dual metric is Nakano seminegative.*

For smooth semipositive line coefficients, Mourougane–Takayama [MT08, Theorem 1.1] imply that the prescribed harmonic metric on

$$R^q\pi_*(K_{X/Y} \otimes L), \qquad 0 \le q \le n,$$

is Nakano semipositive for every proper Kähler submersion. On a complete Kähler base this gives the optimal (m,b) estimates for $1 \le b \le m$. Remark 4.3 gives the detailed metric comparison.

The smooth split-product implication is a consequence of the curvature characterization in [Wat25b] and the fiberwise Bochner identity. The singular split-product result combines scalar weighted estimates with the multiplier-ideal product formula [Dem82, DEL00]. We therefore present these statements as consequences of existing theory. The failure of partial positivity under fiber integration has a close precedent in Inayama's polydisc counterexamples [Ina26, Theorem 1.1];

our compact-fiber examples concern a proper direct image with its prescribed cohomological metric. The top-degree singular result concerns $q = n$ and identifies the actual sheaf and its quotient metric.

Section 7 gives compact-fiber counterexamples to the unrestricted degreewise implication. One occurs on a trivial family in range (i), with a nonsplit metric which is not semipositive; another uses a flat coefficient on a varying elliptic family in range (ii). Thus infinitesimal triviality alone does not replace metric splitting in the original degreewise problem. Section 8 proves Griffiths semipositivity and the minimal extension property for the scalar-factor metrics in the singular positive results.

2. Curvature estimates and weighted cohomology

We use $dV_\omega = \omega^d/d!$ on a Kähler d-fold and the curvature convention $i\Theta_{L,e^{-\varphi}} = i\partial\bar\partial\varphi$. All estimates use the specified form and coefficient metrics. A singular semipositive line metric means $h = e^{-\varphi}$ in local holomorphic frames, with $\varphi \in \mathrm{PSH}$ and $\varphi \not\equiv -\infty$. It is positive and finite almost everywhere and locally bounded below by a positive smooth metric. Put

$$C_{E;r,s} = [i\Theta_{E,h} \otimes \mathrm{Id}, \Lambda_\omega]_{r,s}.$$

For scalar curvature with eigenvalues $\lambda_1, \dots, \lambda_d$, its eigenvalue on $dz^I \wedge d\bar z^J$ is

$$\sum_{i\in I} \lambda_i + \sum_{j\in J} \lambda_j - \sum_{k=1}^{d} \lambda_k. \tag{2.1}$$

Definition 2.1. Let (S, ω) admit a positive Hermitian holomorphic line bundle. In a degree $r = d$, $1 \le s \le d$, or $s = d$, $1 \le r \le d$, a smooth Hermitian bundle (E, h) satisfies the *optimal* (r, s) *estimate* if, for every smooth positive auxiliary line bundle (A, h_A) and every $\bar\partial$-closed $f \in C^\infty_c(S, \Lambda^{r,s}T^*_S \otimes E \otimes A)$, there is a distributional solution $u \in L^2_{r,s-1}(S, E \otimes A)$ with

$$\bar\partial u = f, \qquad \|u\|^2_{hh_A,\omega} \le \int_S \langle B_A^{-1} f, f\rangle_{hh_A,\omega}\, dV_\omega, \quad B_A = [i\Theta_A \otimes \mathrm{Id}_E, \Lambda_\omega]_{r,s}. \tag{2.2}$$

Formula (2.1) gives $B_A > 0$ in these ranges.

Proposition 2.2 (Smooth curvature characterization)**.** *On a complete Kähler manifold admitting a positive Hermitian line bundle, Definition 2.1 is equivalent to $C_{E;r,s} \ge 0$.*

The necessary implication is [Wat25b, Theorems 1.5–1.6], with the fixed-metric definitions in [Wat25b, Definitions 1.3–1.4]. The sufficient implication follows from the next lemma.

For a nonnegative Hermitian operator C, define its extended inverse quadratic form by

$$q_C(f) = \sup_v \{2\,\mathrm{Re}\langle f, v\rangle - \langle Cv, v\rangle\}. \tag{2.3}$$

To see its meaning, diagonalize C at a point, with eigenvalues $\mu_\nu \ge 0$, and write $f = \sum_\nu f_\nu e_\nu$. Maximizing each scalar expression in (2.3) gives

$$q_C(f) = \begin{cases} \displaystyle\sum_{\mu_\nu > 0} \frac{|f_\nu|^2}{\mu_\nu}, & f_\nu = 0 \text{ whenever } \mu_\nu = 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Thus it equals $\langle C^\dagger f, f\rangle$ for $f \perp \ker C$, where $C^\dagger$ is the inverse on $(\ker C)^\perp$ and zero on $\ker C$. This definition also works when the rank of C changes. Directly comparing the expressions inside the supremum gives

$$C \ge B \ge 0 \quad \Longrightarrow \quad q_C \le q_B. \tag{2.4}$$

Lemma 2.3 (Complete Kähler estimate)**.** *Let (S,ω) be complete Kähler, let (E,h) be smooth, and let $C = C_{E;r,s} \geq 0$, where $s \geq 1$. If $f \in L^2_{r,s}(S,E)$, $\bar\partial f = 0$, and $I(f) = \int_S q_C(f)\, dV_\omega < \infty$, then there exists $u \in L^2_{r,s-1}(S,E)$ satisfying $\bar\partial u = f$ and $\|u\|^2 \leq I(f)$.*

Proof. All operators in this proof act between the Hilbert spaces defined by the fixed metrics. Set

$$T = \bar\partial_{\max} : L^2_{r,s-1}(S,E) \longrightarrow L^2_{r,s}(S,E), \qquad S_1 = \bar\partial_{\max} : L^2_{r,s}(S,E) \longrightarrow L^2_{r,s+1}(S,E),$$

with $S_1 = 0$ if $s = d$. They are densely defined and closed, since convergence in their graph norms implies convergence of the corresponding distributional derivatives. Moreover, $\operatorname{im} T \subset \ker S_1$ follows from $\bar\partial^2 = 0$ in distributions. The operator T^* is the Hilbert-space adjoint, whose local expression is the formal $\bar\partial$-adjoint.

The Bochner–Kodaira–Nakano identity gives, for compactly supported smooth w,

$$\|\bar\partial w\|^2 + \|\bar\partial^* w\|^2 = \|D'w\|^2 + \|(D')^* w\|^2 + \int_S \langle Cw, w\rangle\, dV_\omega.$$

We first justify using its inequality for $w \in \operatorname{Dom} S_1 \cap \operatorname{Dom} T^*$. Completeness provides compactly supported Lipschitz cutoffs $0 \leq \chi_\nu \leq 1$ with

$$\chi_\nu \longrightarrow 1 \quad \text{pointwise}, \qquad |d\chi_\nu|_\omega \leq C/\nu.$$

For example, compose the distance from a fixed point with a cutoff that equals one on $[0,\nu]$ and zero on $[2\nu,\infty)$. On a disconnected manifold one first truncates to finitely many components and then makes the same construction on each component. The distributional product formulas are

$$S_1(\chi_\nu w) = \chi_\nu S_1 w + \bar\partial\chi_\nu \wedge w, \qquad T^*(\chi_\nu w) = \chi_\nu T^* w - (\bar\partial\chi_\nu\wedge)^* w.$$

The error terms have norms at most $C\nu^{-1}\|w\|$. Dominated convergence therefore shows that $\chi_\nu w \to w$ in the joint graph norm $\|w\| + \|S_1 w\| + \|T^* w\|$. For a fixed ν, choose finitely many coordinate and bundle charts covering its support. After a smooth partition of unity, convolution of the coefficient distributions approximates $\chi_\nu w$ by compactly supported smooth forms in that same graph norm. This is the local Friedrichs lemma applied simultaneously to the first-order operator (S_1, T^*); its coefficients and the reference metrics are smooth on the compact coordinate sets. A diagonal choice gives smooth w_ν such that

$$w_\nu \to w, \qquad S_1 w_\nu \to S_1 w, \qquad T^* w_\nu \to T^* w \quad \text{in } L^2.$$

Choose a subsequence converging pointwise almost everywhere. Since $C \geq 0$, Fatou's lemma, followed by the smooth Bochner inequality, gives

$$\int_S \langle Cw, w\rangle\, dV_\omega \leq \liminf_\nu \int_S \langle Cw_\nu, w_\nu\rangle\, dV_\omega \leq \|S_1 w\|^2 + \|T^* w\|^2.$$

Finite $I(f)$ implies $f \perp \ker C$ almost everywhere. The diagonal expression for q_C gives the pointwise inequality

$$|\langle f, w\rangle| \leq q_C(f)^{1/2} \langle Cw, w\rangle^{1/2}.$$

After integration and Cauchy–Schwarz, for $w \in \ker S_1 \cap \operatorname{Dom} T^*$ we obtain

$$|(f,w)|^2 \leq I(f) \int_S \langle Cw, w\rangle\, dV_\omega \leq I(f)\|T^* w\|^2.$$

Consequently the rule

$$\ell(T^* w) = (f,w), \qquad w \in \ker S_1 \cap \operatorname{Dom} T^*,$$

defines a bounded functional on this subspace of $L^2_{r,s-1}$: if $T^* w = 0$, the preceding inequality forces $(f,w) = 0$. Its norm is at most $I(f)^{1/2}$. With inner products linear in the first variable, ℓ is antilinear; Hahn–Banach and Riesz give $u \in L^2_{r,s-1}$ with $\ell(v) = (u,v)$ and $\|u\|^2 \leq I(f)$.

It remains to test against all of $\operatorname{Dom} T^*$. For arbitrary $w \in \operatorname{Dom} T^*$, write $w = w_0 + w_\perp$ according to $\ker S_1 \oplus (\ker S_1)^\perp$. Since $\operatorname{im} T \subset \ker S_1$, for every $v \in \operatorname{Dom} T$ one has $(Tv, w_\perp) = 0$.

The definition of the Hilbert adjoint implies $w_\perp \in \operatorname{Dom} T^*$ and $T^* w_\perp = 0$. Thus $w_0 \in \operatorname{Dom} T^* \cap \ker S_1$. Since also $f \in \ker S_1$, we conclude

$$(u, T^* w) = (u, T^* w_0) = (f, w_0) = (f, w).$$

This identity means that $u \in \operatorname{Dom}(T^*)^*$ and $(T^*)^* u = f$. Closedness and dense definition of T give $(T^*)^* = T$, so $\bar\partial u = f$. See also [Dem12, Chapters 4–5]. □

For singular metrics we formulate optimal estimates for finite-energy data.

Definition 2.4. Let E_0 be a holomorphic vector bundle on a smooth Kähler manifold and H a measurable metric, locally bounded below by a positive smooth metric. Define

$$\mathscr{E}(H)(U) = \{s \in H^0(U, E_0) : |s|_H^2 \in L^1_{\mathrm{loc}}(U)\}.$$

The pair $(\mathscr{E}(H), H)$ satisfies the *weighted optimal* (a, b) *estimate* if, for every smooth positive auxiliary line A and every distributionally closed measurable (a, b)-form f with

$$I_A(f) = \int_Y \langle B_A^{-1} f, f\rangle_{Hh_A, \omega_Y}\, dV_Y < \infty,$$

there is a distributional solution $\bar\partial u = f$ with $\|u\|^2_{Hh_A,\omega_Y} \le I_A(f)$. The Dolbeault operators are their maximal distributional realizations.

Finite inverse energy implies local weighted L^2 integrability, since B_A^{-1} has a positive lower bound on every compact set.

Lemma 2.5 (L^2 removability)**.** *Let A be a nowhere-dense analytic subset of a complex manifold. If $u, g \in L^2_{\mathrm{loc}}$ and $\bar\partial u = g$ off A, then this identity holds distributionally everywhere.*

Proof. Work in a coordinate neighborhood $U \subset \mathbb{C}^N$, with a holomorphic bundle frame if necessary. All calculations are coefficientwise, and the smooth reference metrics are equivalent on compact subsets to Euclidean metrics. Let u have type (p, q) and put $k = p + q$. If $q = N$, the equation is automatic, so assume $q < N$ and fix $\tau \in C_c^\infty(U, \Lambda^{N-p,N-q-1})$. Choose a nonzero holomorphic function f on U with $A \cap U \subset \{f = 0\}$ and $|f| < 1$. Set

$$\chi_j = \max\{0, 1 + j^{-1} \log |f|\}.$$

These functions are locally Lipschitz and psh, and satisfy

$$0 \le \chi_j \le 1, \qquad \chi_j = 0 \text{ on } \{|f| \le e^{-j}\}, \qquad \chi_j \longrightarrow 1 \text{ almost everywhere.}$$

For $\eta \in C_c^\infty(U)$, nonnegative smooth regularizations of χ_j and the identity $\Delta(\chi_j^2) = 2|\nabla \chi_j|^2 + 2\chi_j \Delta \chi_j$ give, in the distributional sense,

$$2\int_U \eta^2 |\nabla \chi_j|^2 \le \langle \Delta(\chi_j^2), \eta^2\rangle = \int_U \chi_j^2 \Delta(\eta^2) \le \int_U |\Delta(\eta^2)|.$$

Consequently $\sup_j \|d\chi_j\|_{L^2(K)} < \infty$ for every $K \Subset U$. Also $\chi_j \to 1$ in L^2_{loc}. For every compactly supported smooth real one-form ζ,

$$(d\chi_j, \zeta)_{L^2} = (\chi_j, d^*\zeta)_{L^2} \longrightarrow (1, d^*\zeta)_{L^2} = 0.$$

The local L^2 bound and density of smooth tests imply $d\chi_j \rightharpoonup 0$ in L^2_{loc}.

For fixed j, the support of $\chi_j \tau$ is compact and contained in $\{|f| \ge e^{-j}\} \subset U \setminus A$. Coefficientwise mollification gives forms $\tau_{j,\varepsilon} \in C_c^\infty(U \setminus \{f = 0\})$, with supports in a fixed compact subset of $U \setminus \{f = 0\}$, such that

$$\|\tau_{j,\varepsilon} - \chi_j \tau\|_{W^{1,2}} \longrightarrow 0 \qquad (\varepsilon \to 0).$$

The equation off A gives $\int g \wedge \tau_{j,\varepsilon} = (-1)^{k+1} \int u \wedge \bar\partial \tau_{j,\varepsilon}$. For a compact set containing these supports, Cauchy–Schwarz yields

$$\left|\int g \wedge (\tau_{j,\varepsilon} - \chi_j \tau)\right| + \left|\int u \wedge \bar\partial(\tau_{j,\varepsilon} - \chi_j \tau)\right| \\ \leq C(\|g\|_{L^2} + \|u\|_{L^2}) \|\tau_{j,\varepsilon} - \chi_j \tau\|_{W^{1,2}} \longrightarrow 0.$$

Letting $\varepsilon \to 0$ therefore gives the exact identity

$$\int_U \chi_j g \wedge \tau = (-1)^{k+1} \left(\int_U u \wedge \bar\partial \chi_j \wedge \tau + \int_U \chi_j u \wedge \bar\partial \tau \right).$$

For $K \Subset U$ containing $\operatorname{supp} \tau$, the functional

$$\ell_\tau(\alpha) = \int_U u \wedge \alpha \wedge \tau, \qquad |\ell_\tau(\alpha)| \leq C_\tau \|u\|_{L^2(K)} \|\alpha\|_{L^2(K)}$$

is continuous on $L^2_{0,1}(K)$. Hence $\ell_\tau(\bar\partial \chi_j) \to 0$ by weak convergence. Moreover,

$$\|(\chi_j - 1)u\|^2_{L^2(K)} = \int_K |\chi_j - 1|^2 |u|^2 \longrightarrow 0, \qquad \|(\chi_j - 1)g\|_{L^2(K)} \longrightarrow 0,$$

because $|\chi_j - 1| \leq 1$ and $u, g \in L^2(K)$. Passing to the limit in the preceding identity gives

$$\int_U g \wedge \tau = (-1)^{k+1} \int_U u \wedge \bar\partial \tau \qquad \text{for every } \tau,$$

which is $\bar\partial u = g$ on U. □

Lemma 2.6 (Weighted fiber cohomology)**.** *Let Z be compact Hermitian, and let (M, h_M) have psh local weights. For every p, q, the vector space*

$$V = H^q(Z, \Omega^p_Z \otimes M \otimes \mathcal{I}(h_M))$$

has the positive-definite quotient norm

$$\|v\|^2_{H_V} = \min\{\|\alpha\|^2_{h_M} : \bar\partial \alpha = 0, [\alpha] = v\}. \tag{2.5}$$

The minimum is attained uniquely in $\ker \bar\partial \cap \ker \bar\partial^*$ for the maximal weighted Hilbert complex.

Proof. Let $\mathscr{L}^{p,j}$ consist of germs u such that both u and $\bar\partial u$ are locally $L^2(h_M)$. The complex

$$0 \longrightarrow \Omega^p_Z \otimes M \otimes \mathcal{I}(h_M) \longrightarrow \mathscr{L}^{p,0} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathscr{L}^{p,n} \longrightarrow 0 \tag{2.6}$$

is a fine resolution, as follows. For a smooth partition function ρ, the identity

$$\bar\partial(\rho u) = \rho \bar\partial u + \bar\partial \rho \wedge u$$

shows that multiplication by ρ preserves each graph sheaf: on a relatively compact coordinate set,

$$\|\rho u\|_{h_M} + \|\bar\partial(\rho u)\|_{h_M} \leq C_\rho(\|u\|_{h_M} + \|\bar\partial u\|_{h_M}).$$

Thus smooth partitions of unity act on the sheaves $\mathscr{L}^{p,j}$.

For local exactness, trivialize M on a coordinate ball and write $h_M = e^{-\varphi}$. Let $j \geq 1$ and let $u \in \mathscr{L}^{p,j}$ satisfy $\bar\partial u = 0$. After shrinking to a Euclidean ball B with $\|u\|_{L^2(B, e^{-\varphi})} < \infty$, write

$$u = \sum_{|I|=p} dz^I \wedge u_I, \qquad \bar\partial u_I = 0.$$

For $c > 0$, the scalar Hörmander theorem on B with weight $\varphi + c|z|^2$ gives $(0, j-1)$-forms v_I satisfying

$$\bar\partial v_I = u_I, \qquad \int_B |v_I|^2 e^{-\varphi - c|z|^2} \leq \frac{1}{cj} \int_B |u_I|^2 e^{-\varphi - c|z|^2}.$$

The inverse bound follows from $i\partial\bar\partial(\varphi + c|z|^2) \geq c\, i \sum dz_\ell \wedge d\bar z_\ell$: multiply a scalar $(0,j)$-form by the Euclidean holomorphic volume form to obtain an (n,j)-form, whose curvature operator has lower bound cj. Since $e^{-c|z|^2}$ is bounded above and below by positive constants on B, the form

$$v = (-1)^p \sum_{|I|=p} dz^I \wedge v_I$$

is locally $L^2(h_M)$ and satisfies $\bar\partial v = u$. Therefore $v \in \mathscr{L}^{p,j-1}$. The Euclidean and original smooth form metrics are locally equivalent, so this proves exactness for the prescribed metric. In degree zero, a distributional $\bar\partial$-closed coefficient is holomorphic, and its weighted square integrability is exactly membership in $\mathcal{I}(\varphi)$. This identifies the kernel with $\Omega^p_Z \otimes M \otimes \mathcal{I}(h_M)$. See also [Wat25c, Theorem 5.3(a)].

Write $\mathcal{H}_j = L^2_{p,j}(Z, M, h_M)$ and $d_j = \bar\partial_{\max}$, so

$$\operatorname{Dom} d_j = \{u \in \mathcal{H}_j : \bar\partial u \text{ is represented by an element of } \mathcal{H}_{j+1}\}.$$

These operators are closed. Indeed, if $u_\ell \to u$ in $\mathcal{H}_j$ and $d_j u_\ell \to v$ in $\mathcal{H}_{j+1}$, the local lower bound for h_M gives the same convergences in ordinary L^2_{loc}. Thus $\bar\partial u = v$ in distributions, so $u \in \operatorname{Dom} d_j$ and $d_j u = v$.

To verify dense definition, put $A = \operatorname{Supp}(\mathcal{O}_Z/\mathcal{I}(h_M))$. This is a nowhere-dense analytic subset. On $Z \setminus A$, the weight $e^{-\varphi}$ is locally integrable because $1 \in \mathcal{I}(\varphi)$ there. Hence every compactly supported smooth form in $Z \setminus A$, as well as its derivative, is weighted L^2. Smooth forms are dense for this locally finite weighted measure on relatively compact coordinate sets. Moreover, if χ_ℓ are bounded cutoffs exhausting $Z \setminus A$, then for every $u \in \mathcal{H}_j$,

$$\|(1-\chi_\ell)u\|^2_{h_M} = \int_Z |1-\chi_\ell|^2 |u|^2_{h_M}\, dV_Z \longrightarrow 0,$$

since A has measure zero. A partition of unity and local approximation prove density in all of $\mathcal{H}_j$.

Global sections of the graph sheaves on compact Z are exactly these maximal domains. The fine resolution therefore computes the algebraic, initially unreduced, Hilbert-complex quotient:

$$V = \ker d_q / \operatorname{im} d_{q-1}, \qquad \dim V < \infty.$$

Finiteness follows from coherence of the multiplier ideal and finite-dimensional coherent cohomology on a compact complex manifold. We now prove that this algebraic quotient is separated. For $q \geq 1$, equip $D = \operatorname{Dom} d_{q-1}$ with the graph norm

$$\|v\|^2_D = \|v\|^2_{\mathcal{H}_{q-1}} + \|d_{q-1}v\|^2_{\mathcal{H}_q}, \qquad K = \ker d_q.$$

Both D and K are Banach spaces, and $d_{q-1} : D \to K$ is bounded. Choose a finite-dimensional algebraic complement $W \subset K$ to its range, with the norm induced from K. The bounded surjection

$$D \oplus W \longrightarrow K, \qquad (u, w) \longmapsto d_{q-1}u + w$$

has kernel $\ker d_{q-1} \oplus \{0\}$. The open mapping theorem gives a topological isomorphism

$$(D/\ker d_{q-1}) \oplus W \xrightarrow{\sim} K, \qquad ([v], w) \longmapsto d_{q-1}v + w.$$

The first summand is closed in the domain, and its image is $B_q = \operatorname{im} d_{q-1}$. Therefore B_q is closed in K. Since K is closed in $\mathcal{H}_q$, B_q is also closed in $\mathcal{H}_q$.

Let $\alpha \in K$ represent a class, and let P_{B_q} be the orthogonal projection in $\mathcal{H}_q$. Then

$$\alpha_h = \alpha - P_{B_q}\alpha \in K \cap B_q^\perp, \qquad [\alpha_h] = [\alpha].$$

The definition of the Hilbert adjoint gives

$$B_q^\perp = \ker d^*_{q-1}, \qquad \|\alpha_h + d_{q-1}v\|^2 = \|\alpha_h\|^2 + \|d_{q-1}v\|^2 \quad (v \in \operatorname{Dom} d_{q-1}).$$

This proves existence and uniqueness of the minimum in (2.5). A class has minimum zero only when $\alpha_h = 0$, which is equivalent to its being zero. For $q = 0$, the same argument uses $B_0 = \operatorname{im} d_{-1} = 0$. □

3. Smooth product descent

Proof of Theorem 1.1. Put $V = H^q(Z, \Omega_Z^p \otimes M)$. Product base change and the projection formula give the holomorphic identification $\mathcal{F} = G \otimes_{\mathbb{C}} V$. Choose a basis $v_1, \dots, v_N$ of V and its harmonic representatives $\alpha_1, \dots, \alpha_N$ for the fixed metrics (h_M, ω_Z). If e_G is a local frame of G, the representative of $e_G(y) \otimes v_\nu$ on X_y is $e_G(y) \otimes \alpha_\nu$: the factor $|e_G(y)|^2_{h_G}$ is a positive constant on that fiber, so multiplication by it preserves the minimum-norm representative. Thus the Gram matrix is

$$H_{\nu\bar{\mu}}(y) = |e_G(y)|^2_{h_G} \int_Z \langle \alpha_\nu, \alpha_\mu \rangle_{h_M, \omega_Z} \, dV_Z = |e_G(y)|^2_{h_G} (H_V)_{\nu\bar{\mu}}.$$

This proves the holomorphic isometry

$$(\mathcal{F}, H) = (G, h_G) \otimes (V, H_V), \qquad V = H^q(Z, \Omega_Z^p \otimes M). \tag{3.1}$$

If $V = 0$ there is nothing to prove. Otherwise choose $0 \neq \alpha \in \mathcal{H}^{p,q}(Z, M)$.

The product is complete and admits a positive line bundle, obtained by externally tensoring a positive line on Y with an ample line on Z. Proposition 2.2 therefore gives

$$C_L = [i\Theta_{L,h}, \Lambda_{\omega_X}]_{a+p,b+q} \geq 0.$$

The product cotangent splitting is orthogonal, and $dV_X = dV_Y \, dV_Z$. The curvature and Lefschetz contractions satisfy

$$i\Theta_{L,h} = \mathrm{pr}_Y^* \, i\Theta_{G,h_G} + \mathrm{pr}_Z^* \, i\Theta_{M,h_M}, \qquad \Lambda_X = \Lambda_Y + \Lambda_Z,$$

$$[i\Theta_G \wedge, \Lambda_Z] = 0, \qquad [i\Theta_M \wedge, \Lambda_Y] = 0.$$

Consequently the component $\Lambda^{a,b} T_Y^* \otimes \Lambda^{p,q} T_Z^* \otimes L$ is invariant, and its curvature operator is

$$C_L = C_G \otimes \mathrm{Id} + \mathrm{Id} \otimes C_M, \quad C_G = [i\Theta_G, \Lambda_Y]_{a,b}, \quad C_M = [i\Theta_M, \Lambda_Z]_{p,q}.$$

Fix y and $\xi \in \Lambda^{a,b} T_y^* Y \otimes G_y$. Pointwise nonnegativity of C_L, followed by integration over the fiber, gives

$$0 \leq \int_Z \langle C_L(\xi \otimes \alpha), \xi \otimes \alpha \rangle \, dV_Z = \langle C_G \xi, \xi \rangle \|\alpha\|^2 + |\xi|^2 \int_Z \langle C_M \alpha, \alpha \rangle \, dV_Z.$$

Since $\bar{\partial}_M \alpha = \bar{\partial}_M^* \alpha = 0$, the compact-fiber Bochner identity reads

$$0 = \|\bar{\partial}_M \alpha\|^2 + \|\bar{\partial}_M^* \alpha\|^2 = \|D'_M \alpha\|^2 + \|(D'_M)^* \alpha\|^2 + \int_Z \langle C_M \alpha, \alpha \rangle \, dV_Z.$$

In particular,

$$\int_Z \langle C_M \alpha, \alpha \rangle \, dV_Z = -\|D'_M \alpha\|^2 - \|(D'_M)^* \alpha\|^2.$$

Consequently,

$$\langle C_G \xi, \xi \rangle \|\alpha\|^2 \geq |\xi|^2 (\|D'_M \alpha\|^2 + \|(D'_M)^* \alpha\|^2) \geq 0. \tag{3.2}$$

Dividing (3.2) by $\|\alpha\|^2 > 0$ shows that $C_G \geq 0$ at every y. The constant matrix H_V in the Gram formula implies $i\Theta_{\mathcal{F},H} = i\Theta_{G,h_G} \otimes \mathrm{Id}_V$. Hence $C_{\mathcal{F};a,b} = C_G \otimes \mathrm{Id}_V \geq 0$. For any positive auxiliary line A on Y, the total curvature operator of $\mathcal{F} \otimes A$ is

$$C_{\mathcal{F} \otimes A; a,b} = C_{\mathcal{F};a,b} + B_A \geq B_A > 0.$$

If f is a datum in Definition 2.1, its compact support and smoothness imply $f \in L^2$ and finite inverse energy. Lemma 2.3 and (2.4) give

$$\bar\partial u = f, \qquad \|u\|^2 \le \int_Y q_{C_{\mathcal{F};a,b}+B_A}(f)\, dV_Y \le \int_Y \langle B_A^{-1} f, f\rangle\, dV_Y.$$

□

Remark 3.1. If $C_L \ge 0$ is assumed directly, smooth compact Kähler fibers suffice; projectivity was used only to provide a positive auxiliary line on X.

Proposition 3.2 (Nonsplit metrics in range (ii))**.** *Keep the product geometry and holomorphic line bundle $L = G \boxtimes M$ of Theorem 1.1, but give L an arbitrary smooth metric h. In range (ii), its optimal $(a+p, m+n)$ estimate implies the optimal (a, m) estimate for the prescribed harmonic metric on $R^n\pi_*(\Omega^p_{X/Y} \otimes L)$.*

Proof. Let $V = H^n(Z, \Omega^p_Z \otimes M)$. If $V = 0$ the assertion is empty. Product base change identifies the direct image holomorphically with $G \otimes V$. Choose a basis v_ν and fixed smooth representatives η_ν. On a base coordinate neighborhood U, choose a holomorphic frame of G; the metric h then gives a smooth family of metrics h_y on M. Let Δ_y be the fiberwise $\bar\partial$-Laplacian in degree (p, n). Its kernel has the constant dimension $\dim V$. On fixed Sobolev spaces, the coefficients of Δ_y depend smoothly on y. Choose $\delta > 0$ with $\mathrm{Spec}(\Delta_{y_0}) \cap \{|\zeta| \le 2\delta\} = \{0\}$ at a fixed y_0. After shrinking U, the circle $\Gamma = \{|\zeta| = \delta\}$ remains in the resolvent set and

$$P_y = \frac{1}{2\pi i} \int_\Gamma (\zeta - \Delta_y)^{-1}\, d\zeta$$

is the harmonic projection. Indeed, the spectral projection has locally constant rank, equal at y_0 to $\dim V$, while $\dim \ker \Delta_y = \dim V$ for every y. The resolvent depends smoothly on y between Sobolev spaces: the inverse identity

$$(\zeta - \Delta_y)^{-1} = (\zeta - \Delta_{y_0})^{-1}\big[\mathrm{Id} - (\Delta_y - \Delta_{y_0})(\zeta - \Delta_{y_0})^{-1}\big]^{-1}$$

is a locally convergent operator Neumann series, uniformly on Γ. Elliptic regularity and Sobolev embedding give

$$y \longmapsto P_y\eta_\nu \in C^\infty(U, H^s(Z)) \quad \text{for every } s, \qquad (y, z) \longmapsto (P_y\eta_\nu)(z) \in C^\infty(U \times Z).$$

Set $\alpha_\nu(y) = P_y\eta_\nu$. If G_y is the Green operator, the top-antiholomorphic-degree Hodge identity gives

$$\eta_\nu - \alpha_\nu(y) = \Delta_y G_y \eta_\nu = \bar\partial_Z \bar\partial^*_{h_y} G_y \eta_\nu.$$

Thus $\alpha_\nu(y)$ represents v_ν, and

$$H_{\nu\bar\mu}(y) = \int_Z \langle \alpha_\nu(y), \alpha_\mu(y)\rangle_{h_y,\omega_Z}\, dV_Z$$

is the smooth Gram matrix of the prescribed metric, in the frame of G just chosen. A change of that frame rescales h_y by a constant on each fiber and preserves its harmonic subspace.

For $0 \le j \le m$, define the harmonic lift and the cohomology projection by

$$J_j v = \sum_\nu \pi^* v_\nu \wedge \alpha_\nu(y), \qquad \Pi_j W = [W_{(a,j),(p,n)}]_{\bar\partial_Z}.$$

Here W has total bidegree $(a + p, j + n)$, and the subscript selects horizontal degree (a, j) and vertical degree (p, n). Every vertical (p, n)-form is $\bar\partial_Z$-closed. In a horizontal orthonormal form basis, apply the minimum-norm property separately to each vertical coefficient $w_{I,J}$. This gives

$$|\Pi_j W(y)|^2_H = \sum_{I,J} \|[w_{I,J}]\|^2_{H_y} \le \sum_{I,J} \|w_{I,J}\|^2_{L^2(Z,h_y)} \le \int_Z |W(y, z)|^2_{h,\omega_X}\, dV_Z.$$

For a harmonic lift the first inequality is an equality and the other product components vanish. Integrating over Y yields

$$\Pi_j J_j = \mathrm{Id}, \qquad \|J_j v\|_X = \|v\|_Y, \qquad \|\Pi_j W\|_Y \le \|W\|_X. \tag{3.3}$$

These identities remain valid after twisting by a base line A, since its norm is constant on each fiber.

To compute the projection, choose the fixed Serre-dual basis $\beta^\nu \in H^0(Z, \Omega_Z^{n-p} \otimes M^{-1})$, normalized by

$$\int_Z \eta_\mu \wedge \beta^\nu = \delta_\mu^\nu.$$

In local holomorphic frames, write the selected component as $\sum_{I,J} dt^I \wedge d\bar{t}^J \wedge w_{I,J}$. Then

$$\Pi_j W = \sum_{I,J,\nu} \left(\int_Z w_{I,J}(y,z) \wedge \beta^\nu(z) \right) dt^I \wedge d\bar{t}^J \otimes v_\nu.$$

Thus Π_j is independent of the coefficient metric. For $0 \le j < m$, the component formula is

$$(\bar\partial_X W)_{(a,j+1),(p,n)} = \bar\partial_Y W_{(a,j),(p,n)} + (-1)^{a+j+1} \bar\partial_Z W_{(a,j+1),(p,n-1)}$$

For a smooth vertical $(p, n-1)$ coefficient γ,

$$\int_Z \bar\partial_Z \gamma \wedge \beta^\nu = \int_Z \bar\partial_Z (\gamma \wedge \beta^\nu) = 0, \qquad \bar\partial_Z \beta^\nu = 0.$$

Consequently the coefficients of the preceding component formula satisfy

$$\int_Z (\bar\partial_X W)_{(a,j+1),(p,n)} \wedge \beta^\nu = \bar\partial_Y \int_Z W_{(a,j),(p,n)} \wedge \beta^\nu.$$

Equivalently,

$$\Pi_{j+1} \bar\partial_X W = \bar\partial_Y \Pi_j W. \tag{3.4}$$

The same identity holds for W in the maximal domain. For every compactly supported base test form τ, the form $\pi^*\tau \wedge \beta^\nu$ has compact support on $U \times Z$. Distributional differentiation and fiber integration give

$$\mathrm{pr}_{Y*}(\bar\partial_Z \gamma \wedge \beta^\nu) = 0, \qquad \mathrm{pr}_{Y*}(\bar\partial_Y \eta \wedge \beta^\nu) = \bar\partial_Y \,\mathrm{pr}_{Y*}(\eta \wedge \beta^\nu)$$

when paired with τ, for any distributional vertical coefficients γ, η of the indicated degrees. These are the same integration-by-parts identities, since $\bar\partial_Z(\pi^*\tau \wedge \beta^\nu) = 0$. In particular neither summand in the component formula has to be separately L^2. The bounded projection estimate ensures that both sides of (3.4), whenever the left side is defined from $\bar\partial_X W \in L^2$, represent the claimed distributional derivative of an L^2 form.

Let f be a compactly supported smooth (a, m) datum and let $A > 0$. The lift $F = J_m f$ is smooth and supported in $\mathrm{supp}\, f \times Z$, hence is L^2 with compact support. It has total antiholomorphic degree $m + n$, so $\bar\partial_X F = 0$ for degree reasons. For

$$B_X = [i\pi^*\Theta_A, \Lambda_X]_{a+p,m+n}, \qquad B_Y = [i\Theta_A, \Lambda_Y]_{a,m},$$

Choose horizontal unitary covectors with $i\Theta_A = i\sum_{i=1}^m \lambda_i \, dt_i \wedge d\bar{t}_i$, $\lambda_i > 0$, and vertical unitary covectors on Z. The pulled-back curvature has eigenvalues $\lambda_1, \dots, \lambda_m, 0, \dots, 0$. On a total $(a+p, m+n)$ component with horizontal holomorphic index set I and vertical holomorphic index set K, formula (2.1) gives

$$B_X\big|_{I,K} = \left(\sum_{i \in I} \lambda_i \right) \mathrm{Id}, \qquad |I| + |K| = a + p.$$

Therefore $B_X \geq 0$; its possible zero eigenspaces have $I = \varnothing$. The lift $J_m f$ lies in the block $|I| = a \geq 1$, $|K| = p$, on which the eigenvalue is exactly that of B_Y on the corresponding horizontal component. Hence

$$B_X J_m = J_m B_Y, \qquad J_m f \perp \ker B_X, \qquad B_X^{\dagger} J_m f = J_m(B_Y^{-1} f).$$

The lift isometry and its polarized identity now give the exact inverse-energy formula

$$\int_X q_{B_X}(J_m f)\, dV_X = \int_X \langle J_m(B_Y^{-1} f), J_m f\rangle\, dV_X = \int_Y \langle B_Y^{-1} f, f\rangle\, dV_Y.$$

The hypothesis and Proposition 2.2 give $C_L \geq 0$. The curvature of $L \otimes \pi^* A$ is represented by $C_L + B_X \geq B_X$. Equations (2.3)–(2.4) give

$$\int_X q_{C_L + B_X}(J_m f)\, dV_X \leq \int_X q_{B_X}(J_m f)\, dV_X < \infty.$$

The complete-manifold estimate supplies U with $\bar\partial_X U = J_m f$ and $\|U\|^2 \leq \int_Y \langle B_Y^{-1} f, f\rangle\, dV_Y$. Set $u = \Pi_{m-1} U$. Finally,

$$\bar\partial_Y u = \Pi_m \bar\partial_X U = \Pi_m J_m f = f, \qquad \|u\|^2 \leq \|U\|^2 \leq \int_Y \langle B_Y^{-1} f, f\rangle\, dV_Y,$$

by (3.3)–(3.4). □

Corollary 3.3 (Descent from a dense metrically split locus)**.** *Let $\pi : X \to Y$ be a proper holomorphic submersion, with X and Y complete Kähler and admitting positive line bundles. Assume that the dimensions $h^q(X_y, \Omega^p_{X_y} \otimes L_y)$ are constant. Suppose that, on a dense open subset Y°, there are local product charts in which*

$$\omega_X = \pi^* \omega_Y + \omega_Z, \qquad (L, h) = (G_U, h_{G_U}) \boxtimes (M_U, h_{M_U}),$$

with fixed fiber data. In either range (1.1), the upstairs optimal estimate implies the downstairs optimal estimate.

Proof. The upstairs hypothesis and Proposition 2.2 give $C_L \geq 0$ on X. Fix a product chart over $U \subset Y^\circ$. If the common cohomology rank is zero, the conclusion is immediate. Otherwise choose a nonzero harmonic fiber class α. For a horizontal tensor ξ, the same pointwise splitting and compact-fiber integration used in (3.2) give

$$\langle C_{G_U;a,b}\xi, \xi\rangle \|\alpha\|^2 \geq |\xi|^2 (\|D'_{M_U}\alpha\|^2 + \|(D'_{M_U})^*\alpha\|^2) \geq 0.$$

The local Gram matrix is $h_{G_U} H_{V_U}$ with H_{V_U} constant, so

$$C_{\mathcal{F};a,b} = [i\Theta_{\mathcal{F},H}, \Lambda_Y]_{a,b} = C_{G_U;a,b} \otimes \mathrm{Id}_{V_U} \geq 0 \quad \text{on } U.$$

To pass across $Y \setminus Y^\circ$, constant cohomology dimension and Grauert's theorem give local freeness and base change for $\mathcal{F}$. In a local smooth trivialization of the proper submersion, the fiber Laplacians are smooth elliptic families. Their kernels have constant dimension; the spectral projection construction above therefore gives smooth harmonic representatives of local frames. Their Gram entries are integrals over a fixed compact smooth fiber of smoothly varying coefficients, so $H_{\nu\bar\mu} \in C^\infty(Y)$. In particular $C_{\mathcal{F};a,b}$ is continuous on all of Y.

Fix $y_0 \in Y$, a sequence $y_\ell \in Y^\circ$ with $y_\ell \to y_0$, and $\xi_0 \in \Lambda^{a,b} T^*_{y_0} Y \otimes \mathcal{F}_{y_0}$. Choose a local smooth section ξ with $\xi(y_0) = \xi_0$. Then

$$\langle C_{\mathcal{F};a,b}(y_0)\xi_0, \xi_0\rangle = \lim_{\ell\to\infty} \langle C_{\mathcal{F};a,b}(y_\ell)\xi(y_\ell), \xi(y_\ell)\rangle \geq 0.$$

Thus $C_{\mathcal{F};a,b} \geq 0$ everywhere. For each positive auxiliary line A on the complete base, $C_{\mathcal{F}\otimes A;a,b} \geq B_A > 0$. Lemma 2.3 and (2.4) give the claimed optimal estimate. □

Infinitesimal triviality supplies dense holomorphic product charts, but not the two metric splittings in this corollary. The counterexample in Corollary 7.2 is already a globally trivial family.

4. Top direct images and the smooth adjoint case

Let $\pi : X \to Y$ be a proper surjective holomorphic submersion of relative dimension n between Kähler manifolds. A smooth Hermitian bundle (E, h) is *dual Nakano semipositive* if (E^*, h^*) is Nakano seminegative. The submersion is *infinitesimally trivial* if its Kodaira–Spencer maps

$$\rho_y : T_y^{1,0}Y \longrightarrow H^1(X_y, T_{X_y})$$

vanish for every y. In this section the dimension $h^n(X_y, E_y)$ is assumed locally constant.

Theorem 1.4 is the metric Serre-dual form of the negative-adjoint mechanism in Liu–Yang [LY14, Theorem 4.1]. We give the metric identification and the dense-locus argument explicitly.

Lemma 4.1 (Isometric Serre duality)**.** *For a compact complex n-fold Z and a smooth Hermitian bundle (E, h), the quotient norm on $H^n(Z, E)$ is independent of the Hermitian metric on Z. With the canonical L^2 norm on $H^0(Z, K_Z \otimes E^*)$, Serre duality is an isometry. In the family,*

$$(R^n\pi_*E, H) \simeq \big(\pi_*(K_{X/Y} \otimes E^*), G\big)^* \tag{4.1}$$

as smooth Hermitian holomorphic bundles.

Proof. The ambient Hilbert norm. In a holomorphic coordinate chart put $d\lambda_z = i^{n^2} dz_1 \wedge \cdots \wedge dz_n \wedge d\bar z_1 \wedge \cdots \wedge d\bar z_n$. For $a = a_0\, d\bar z_1 \wedge \cdots \wedge d\bar z_n$ and $\omega = i \sum g_{j\bar k} dz_j \wedge d\bar z_k$,

$$|a|^2_{h,\omega} = (\det g)^{-1} |a_0|^2_h, \qquad dV_\omega = (\det g) d\lambda_z, \qquad |a|^2_{h,\omega} dV_\omega = |a_0|^2_h d\lambda_z.$$

The same identity, with h^* in place of h, holds for $s = s_0\, dz_1 \wedge \cdots \wedge dz_n$. Thus both Hilbert norms are independent of ω. The subspace

$$\mathcal{B} = \overline{\bar\partial A^{0,n-1}(Z, E)}^{\,L^2_{0,n}}$$

is the closure of the same set in this fixed Hilbert norm. For a smooth representative a, compact Hodge theory gives $a = a_h + \bar\partial b$ with $a_h \perp \mathcal{B}$, and therefore

$$\inf_{c \in \mathcal{B}} \|a + c\|^2 = \|a_h\|^2, \qquad \|a_h + c\|^2 = \|a_h\|^2 + \|c\|^2.$$

This proves the asserted independence of the quotient norm.

The duality map and equality of norms. The exterior-product pairing defines an antilinear isometry J by

$$s \wedge a = \langle s, Ja\rangle_{h^*,\omega} dV_\omega, \qquad J : \Lambda^{0,n} T_Z^* \otimes E \longrightarrow \Lambda^{n,0} T_Z^* \otimes E^*.$$

The fixed unit phase from the exterior ordering is included in J. With the Hermitian products linear in their first entry, one has

$$(Ja, Jc)_{L^2} = (c, a)_{L^2}, \qquad \int_Z Ja \wedge \bar\partial b = (\bar\partial b, a)_{L^2}.$$

For smooth a and $b \in A^{0,n-1}(Z, E)$, Stokes gives

$$\int_Z Ja \wedge \bar\partial b = (-1)^{n+1} \int_Z \bar\partial(Ja) \wedge b.$$

Since b is arbitrary, the nondegenerate test-form pairing implies

$$a \perp \bar\partial A^{0,n-1}(Z, E) \iff \int_Z Ja \wedge \bar\partial b = 0 \quad \text{for all } b \iff \bar\partial(Ja) = 0.$$

Every $(0, n)$-form is $\bar\partial$-closed. Consequently J identifies harmonic $(0, n)$-forms antiunitarily with $H^0(Z, K_Z \otimes E^*)$. For the minimum representative a_h, the integral $\int_Z s \wedge a$ depends only on $[a]$, because $\bar\partial s = 0$. Moreover,

$$\left|\int_Z s \wedge a\right| = \left|\int_Z s \wedge a_h\right| \le \|s\|_{L^2} \|a_h\|_{L^2}, \qquad \int_Z Ja_h \wedge a_h = \|a_h\|^2_{L^2}.$$

The upper bound is thus attained, and

$$\|[a]\|_H^2 = \sup_{s\neq 0} \frac{|\int_Z s\wedge a|^2}{\|s\|_G^2}; \tag{4.2}$$

For the zero class both sides are zero; otherwise use $s = Ja_h$.

Holomorphic and smooth dependence on the base. Constant fiber dimension and Grauert's theorem [Gra60] give local freeness and base change. The Serre pairing is holomorphic by relative duality [RRV71]: for a relative Dolbeault representative, its base $\bar\partial$ derivative differs from the representative of the cohomological derivative by a fiberwise exact term, whose integral against a holomorphic top form vanishes. Let $s_1,\dots,s_r$ be a local holomorphic frame of $\pi_*(K_{X/Y}\otimes E^*)$ and let $e_1,\dots,e_r$ be its Serre-dual frame. Fiberwise Serre duality gives the constant rank of the first bundle, and

$$G_{\alpha\bar\beta}(y) = \int_{X_y} \langle s_\alpha(y), s_\beta(y)\rangle_{h^*,\omega_y} dV_{\omega_y}, \qquad H(y) = (G(y)^{-1})^{\mathsf T}.$$

To justify smoothness of G, use a local C^∞ trivialization $\pi^{-1}(U)\simeq U\times X_{y_0}$ and a finite partition of unity on its compact fibers. Each matrix entry is then a finite sum $\int_{X_{y_0}} a_{\alpha\bar\beta}(y,z)\,dV_0(z)$ with smooth integrand. On relatively compact $U'\Subset U$, every derivative of the integrand is bounded on $\overline{U'}\times X_{y_0}$, so

$$\partial_y^\gamma G_{\alpha\bar\beta}(y) = \int_{X_{y_0}} \partial_y^\gamma a_{\alpha\bar\beta}(y,z)\,dV_0(z).$$

The matrix $G(y)$ is positive definite, hence its inverse and H are smooth. Formula (4.2) identifies this dual metric with the prescribed harmonic metric. □

Lemma 4.2 (Dense product locus)**.** *An infinitesimally trivial proper holomorphic submersion is holomorphically locally a product over a dense open $Y^\circ\subset Y$. Locally on Y, the complement can be chosen analytic.*

Proof. Work over a connected coordinate neighborhood and remove the analytic locus where $h^1(X_y,T_{X_y})$ exceeds its generic value. Grauert's theorem gives, on the remaining dense open set,

$$\mathcal T = R^1\pi_* T_{X/Y} \text{ locally free}, \qquad \mathcal T\otimes\mathbb C(y)\simeq H^1(X_y,T_{X_y}).$$

These local dense open sets cover a dense open $Y^\circ\subset Y$. Apply π_* to

$$0\longrightarrow T_{X/Y}\longrightarrow T_X\xrightarrow{d\pi}\pi^*T_Y\longrightarrow 0.$$

The resulting exact sequence contains

$$\pi_*T_X\longrightarrow\pi_*\pi^*T_Y\xrightarrow{\delta}\mathcal T.$$

For $\xi_j=\partial/\partial t_j$ regarded as a section of $\pi_*\pi^*T_Y$, base change identifies

$$\delta(\xi_j)(y)=\rho_y(\xi_j(y))=0.$$

Write $\delta(\xi_j)=\sum_\alpha f_{j\alpha}e_\alpha$ in a local frame of $\mathcal T$. Thus $f_{j\alpha}(y)=0$ for every y, and $\delta(\xi_j)=0$ as a section. Exactness, followed by one shrinking for the finitely many indices j, gives

$$V_j\in H^0(\pi^{-1}(U),T_X), \qquad d\pi(V_j)=\xi_j.$$

Center the coordinates at $y_0=0$ and let Φ_j^ζ be the local holomorphic flow of V_j. Compactness of X_0 gives a common $\varepsilon>0$ on which all successive flows below are defined near this fiber for $|t_j|<\varepsilon$. Their base coordinates satisfy

$$\frac{d}{d\zeta}\pi(\Phi_j^\zeta(x))=e_j, \qquad \pi(\Phi_j^\zeta(x))=\pi(x)+\zeta e_j.$$

Consequently

$$F(t,z)=\Phi_m^{t_m}\circ\cdots\circ\Phi_1^{t_1}(z),\qquad F:\Delta_\varepsilon^m\times X_0\longrightarrow X,\qquad \pi F(t,z)=t.$$

At $(0,z)$, dF identifies the vertical tangent with T_zX_0 and sends the coordinate directions to $V_j(z)$, so it is invertible. Compactness allows a common shrinking on which it stays invertible. For fixed t, the composition of flows is injective; different t have different images under π. Thus F is a biholomorphism onto an open W containing X_0, with inverse

$$x\longmapsto\left(t(x),\Phi_1^{-t_1(x)}\circ\cdots\circ\Phi_m^{-t_m(x)}(x)\right).$$

These two maps are inverse. The relatively closed set $C=\pi^{-1}(\Delta_\varepsilon^m)\setminus W$ has closed image $\pi(C)$ by properness, and $0\notin\pi(C)$. Choose $U'\ni 0$ disjoint from $\pi(C)$. Then

$$\pi^{-1}(U')\subset W,\qquad F|_{U'\times X_0}:U'\times X_0\xrightarrow{\sim}\pi^{-1}(U').$$

This proves local product structure at every point of Y°. □

Proof of Theorem 1.4. Put $V=E^*$. By Lemma 4.2, it suffices first to work on a product chart $U\times Z$ over Y°. Let P_y be the orthogonal projection onto $H^0(Z,K_Z\otimes V_y)$ and let $\nabla_i,\nabla_{\bar j}$ be the horizontal Chern derivatives of V. The canonical density of Lemma 4.1 permits integration with a fixed fiber metric. Therefore the metric derivative contains no derivative of a fiber volume form:

$$\partial_i(s,t)_{L^2}=(\nabla_i s,t)_{L^2}+(s,\nabla_{\bar i}t)_{L^2}.$$

For s,t in the direct image, orthogonality gives $(\nabla_i s,t)=(P\nabla_i s,t)$. Moreover, a holomorphic direct-image section is a holomorphic relative top form on $U\times Z$, so $\nabla_{\bar j}s=0$. Thus the projected connection has the correct metric compatibility and $(0,1)$ part, and is the Chern connection:

$$D_i^G=P\nabla_i,\qquad B_i=(1-P)\nabla_i|_{\operatorname{im}P}.$$

The operator B_i is tensorial: for a holomorphic function f on U,

$$(1-P)\nabla_i(fs)=(1-P)((\partial_i f)s+f\nabla_i s)=fB_i s.$$

We now compute its curvature contribution rather than assume a subbundle formula. The result is

$$\langle R^G_{i\bar j}s,t\rangle_G=\int_Z\langle R^V_{i\bar j}s,t\rangle_{h^*,\omega_Z}\,dV_Z-(B_is,B_jt)_{L^2}.\tag{4.3}$$

Choose a local holomorphic frame normal for G at y_0, and take s,t with constant coefficients in this frame. The normality conditions are $G(y_0)=\mathrm{Id}$ and $\partial_iG(y_0)=0$, hence $P\nabla_i s(y_0)=0$. Since $R^V_{i\bar j}=[\nabla_i,\nabla_{\bar j}]$ and $\nabla_{\bar j}s=0$, differentiating the preceding metric identity gives

$$\begin{aligned}\bar\partial_j\partial_i(s,t)_{L^2}&=(\nabla_{\bar j}\nabla_i s,t)_{L^2}+(\nabla_i s,\nabla_j t)_{L^2}\\&=-\int_Z\langle R^V_{i\bar j}s,t\rangle_{h^*,\omega_Z}\,dV_Z+(\nabla_i s,\nabla_j t)_{L^2}.\end{aligned}$$

At y_0, $\nabla_i s=B_i s$ and $\langle R^G_{i\bar j}s,t\rangle=-\bar\partial_j\partial_i(s,t)_{L^2}$. Substitution proves (4.3). For arbitrary $s_1,\dots,s_m$,

$$\sum_{i,j}\langle R^G_{i\bar j}s_i,s_j\rangle_G=\int_Z\sum_{i,j}\langle R^V_{i\bar j}s_i,s_j\rangle_{h^*,\omega_Z}\,dV_Z-\Big\|\sum_i B_is_i\Big\|^2\le 0.$$

For the first term, write $s_i(z)=v_i(z)\,dz_1\wedge\cdots\wedge dz_n$ in a fiber coordinate chart. Nakano seminegativity of V applied to $\sum_i\partial_{t_i}\otimes v_i(z)$ gives

$$\sum_{i,j}\langle R^V_{i\bar j}v_i(z),v_j(z)\rangle_{h^*}\le 0.$$

Multiplication by the positive canonical density and integration prove the asserted sign. Thus G is Nakano seminegative on Y°.

To pass across its complement, choose any $y_0 \in Y$ and extend the entries of a tensor in $T^{1,0}_{y_0}Y \otimes G_{y_0}$ smoothly to $s_1(y), \dots, s_m(y)$ in local frames. Lemma 4.1 makes the function

$$Q(y) = \sum_{i,j} \langle R^G_{i\bar{j}}(y) s_i(y), s_j(y) \rangle_G$$

continuous. For $y_\nu \in Y^\circ$ tending to y_0, $Q(y_0) = \lim_\nu Q(y_\nu) \leq 0$. The isometry (4.1) gives the asserted dual Nakano semipositivity of H. □

Remark 4.3 (Smooth semipositive adjoints and the prescribed metric). Let $\pi : X \to Y$ be a proper Kähler holomorphic submersion and let (L, h) be a smooth semipositive line bundle on X. For every $0 \leq q \leq n$, the ordinary fiberwise harmonic metric on

$$\mathcal{E}_q = R^q\pi_*(K_{X/Y} \otimes L)$$

is Nakano semipositive. This is a consequence of Mourougane–Takayama [MT08, Theorem 1.1], with the following explicit comparison of metrics. Connected fibers may be assumed: locally on the base, disconnected fibers give an orthogonal direct sum of the connected cases.

Write $m = \dim Y$, $c_d = i^{d^2}$ and $\omega_y = \omega_X|_{X_y}$. In the notation of [MT08, Corollary 4.4], take the relative injection

$$S_\omega : \mathcal{E}_q \longrightarrow \pi_*(\Omega^{n-q}_{X/Y} \otimes L).$$

For a local holomorphic section s, put $\sigma = S_\omega s$. The base-change assertion in [MT08, Lemma 4.9] is

$$\mathcal{E}_q \otimes \mathbb{C}(y) \xrightarrow{\sim} H^q(X_y, K_{X_y} \otimes L_y).$$

Thus every fiber class is the value of a local holomorphic section. Corollary 4.4 and Lemma 4.7(1) of [MT08] give

$$\bar\partial_y \sigma_y = 0, \qquad D'_{h_y}\sigma_y = 0, \qquad s_y = \left[\frac{c_{n+m-q}}{q!}\omega_y^q \wedge \sigma_y\right]. \tag{4.4}$$

Work over a Stein coordinate ball on the base. The Leray spectral sequence and Cartan's theorem B identify sections of $\mathcal{E}_q$ with degree-q cohomology on its inverse image. If ρ is a Dolbeault representative of s, the identity $(c_{n+m-q}/q!)L_\omega S_\omega = \mathrm{Id}$ says, after the relative canonical bundle is identified using $dt = dt_1 \wedge \cdots \wedge dt_m$,

$$(\rho - c_{n+m-q}\omega_X^q \wedge \sigma/q!) \wedge dt = \bar\partial(b \wedge dt).$$

Restricting the relative coefficients to X_y kills all horizontal differentials and gives

$$\rho_y - c_{n+m-q}\omega_y^q \wedge \sigma_y/q! = \bar\partial_y b_y.$$

Set $\alpha_y = c_{n+m-q}\omega_y^q \wedge \sigma_y/q!$. The fiber star identity [MT08, Corollary 2.2] yields

$$*_y \alpha_y = \frac{c_{n+m-q}}{c_{n-q}}\sigma_y.$$

Since ω_y is Kähler, (4.4) gives

$$\bar\partial_y \alpha_y = 0, \qquad \bar\partial^*_y \alpha_y = - *_y D'_{h_y}(*_y \alpha_y) = 0.$$

To check the minimizing property explicitly, every representative of the same class has the form $\alpha_y + \bar\partial_y v$. On the compact fiber, integration by parts gives

$$(\alpha_y, \bar\partial_y v)_{L^2} = (\bar\partial^*_y \alpha_y, v)_{L^2} = 0, \qquad \|\alpha_y + \bar\partial_y v\|^2 = \|\alpha_y\|^2 + \|\bar\partial_y v\|^2.$$

The identity extends to the maximal domain by graph approximation. Thus α_y is the ordinary harmonic, minimum-norm representative. For the metric of [MT08, Definition 5.1], the pointwise Lefschetz identity gives

$$g_{\mathrm{MT},y}(s_y, s_y) = \frac{c_{n-q}}{q!}\int_{X_y} \omega_y^q \wedge \sigma_y \wedge h_y\overline{\sigma_y} = \int_{X_y} |\sigma_y|^2_{h_y,\omega_y}\, dV_y.$$

The star is an isometry and $|c_{n+m-q}/c_{n-q}| = 1$, so

$$H_y(s_y, s_y) = \|\alpha_y\|^2_{L^2} = \|\sigma_y\|^2_{L^2} = g_{\mathrm{MT},y}(s_y, s_y). \tag{4.5}$$

This equality holds on every fiber vector. Polarization therefore gives $H_y(s_y, t_y) = g_{\mathrm{MT},y}(s_y, t_y)$ for all s_y, t_y.

Mourougane–Takayama's curvature theorem therefore applies to the prescribed H. For a complete Kähler base and $1 \le b \le m$, Nakano semipositivity implies

$$[i\Theta_{\mathcal{E}_q,H}, \Lambda_{\omega_Y}]_{m,b} \ge 0.$$

Indeed, write in unitary coordinates $u = \sum_{|J|=b} u_J\, dz_1 \wedge \cdots \wedge dz_m \wedge d\bar{z}_J$, and let u_{iK} be its signed alternating coefficients, zero if $i \in K$. Then

$$\langle [i\Theta_{\mathcal{E}_q,H}, \Lambda]u, u\rangle = \sum_{|K|=b-1} \sum_{i,j} \langle R_{i\bar{j}} u_{iK}, u_{jK}\rangle_H \ge 0.$$

Each inner sum is the Nakano form evaluated on $\sum_i \partial_{z_i} \otimes u_{iK}$. After twisting by a positive auxiliary line A, its commutator is

$$C_{\mathcal{E}_q\otimes A;m,b} = [i\Theta_{\mathcal{E}_q,H}, \Lambda]_{m,b} + B_A \ge B_A > 0.$$

For every datum in Definition 2.1, Lemma 2.3 and inverse order (2.4) give

$$\bar\partial u = f, \qquad \|u\|^2_{Hh_A} \le \int_Y \langle B_A^{-1} f, f\rangle_{Hh_A}\, dV_Y, \quad B_A = [i\Theta_A \otimes \mathrm{Id}, \Lambda_{\omega_Y}]_{m,b}.$$

This proves general descent in range (i) for smooth semipositive L, including $q > 0$. The same comparison works for a smooth Nakano semipositive vector coefficient.

5. Singular product metrics

Lemma 5.1 (Scalar estimates in the two extremal ranges)**.** *Let (G, h_G) be a holomorphic line bundle with psh local weights. If (Y, ω_Y) is complete Kähler of dimension m, then $(G\otimes\mathcal{I}(h_G), h_G)$ satisfies the weighted optimal (m, b) estimate for $1 \le b \le m$ and the weighted optimal (a, m) estimate for $1 \le a \le m$. The second assertion holds on any Kähler manifold.*

Proof. Fix a smooth positive auxiliary metric h_A on a holomorphic line bundle A, and write $E = G \otimes A$ and $h = h_G h_A$. For the first assertion, choose a smooth reference metric $h_{G,0}$ and define the globally defined function χ by

$$h_G = h_{G,0} e^{-\chi}, \qquad h_0 = h_{G,0} h_A, \qquad h = h_0 e^{-\chi}.$$

Locally χ is a psh function plus a smooth function. Its distributional complex Hessian is therefore a current of order zero, whose singular part is nonnegative. Denote by

$$R = i\Theta_{E,h_0} + (i\partial\bar\partial\chi)_{\mathrm{ac}} = i\Theta_{A,h_A} + (i\Theta_{G,h_G})_{\mathrm{ac}}$$

the absolutely continuous curvature form. It has measurable coefficients and satisfies $R \ge i\Theta_{A,h_A}$ almost everywhere. On (m, b)-forms, put

$$C = [R, \Lambda_Y]_{m,b}, \qquad B_A = [i\Theta_{A,h_A}, \Lambda_Y]_{m,b}.$$

The eigenvalue formula (2.1) gives $C \ge B_A > 0$, and hence

$$\langle C^{-1} f, f\rangle_h \le \langle B_A^{-1} f, f\rangle_h \quad \text{almost everywhere.}$$

If the right-hand side has finite integral, then on each compact $K \subset Y$ there is $c_K > 0$ with

$$B_A^{-1} \ge c_K\, \mathrm{Id}, \qquad \int_K |f|^2_h\, dV_Y \le c_K^{-1} I_A(f).$$

In particular, f has locally L^2 coefficients also for the smooth reference metric, since h is locally bounded below.

Apply [Dem82, Section 5, Theorem 5.1] to the smooth line bundle (E, h_0) and the singular weight χ. The input of that theorem is precisely a distributionally closed (m, b)-form with locally L^2 coefficients and finite inverse energy for the absolutely continuous curvature. Its conclusion, in the present notation, is

$$\bar\partial u = f, \qquad \int_Y |u|_h^2\, dV_Y \le \int_Y \langle C^{-1} f, f\rangle_h\, dV_Y \le I_A(f).$$

This proves the first estimate for the specified ω_Y.

For the second assertion, set $k = h^{-1}$ on E^* and $r = m - a$. The metric k is positive and finite almost everywhere and locally bounded above. Define

$$B = [i\Theta_{A,h_A}, \Lambda_Y]_{a,m}, \qquad Q = [-i\Theta_{A,h_A}, \Lambda_Y]_{r,0}.$$

If $i\Theta_{A,h_A}$ is diagonal in a unitary coframe, with eigenvalues $\lambda_1, \dots, \lambda_m > 0$, then

$$B(dz^I \wedge d\bar z^1 \wedge \cdots \wedge d\bar z^m) = \beta_I\, dz^I \wedge d\bar z^1 \wedge \cdots \wedge d\bar z^m, \qquad \beta_I = \sum_{i\in I} \lambda_i, \quad |I| = a,$$

whereas $Q(dz^{I^c}) = \beta_I dz^{I^c}$. In particular, both operators are positive. We prove the compactly supported dual-test inequality

$$\int_Y \langle Qw, w\rangle_k\, dV_Y \le \|\bar\partial w\|_k^2, \qquad w \in C_c^\infty(Y, \Lambda^{r,0} T_Y^* \otimes E^*). \tag{5.1}$$

All terms are finite because k is locally bounded above.

Choose a coordinate ball with holomorphic frames and write $h_G = e^{-\varphi}$ and $h_A = e^{-\psi}$, where φ is psh and ψ is smooth. On a smaller ball, convolution with radial regularizing kernels gives smooth psh functions $\varphi_\varepsilon \downarrow \varphi$. For a compactly supported smooth $(r, 0)$-form v in this ball, set

$$k_\varepsilon = e^{\varphi_\varepsilon + \psi}, \qquad s_\varepsilon = e^{(\varphi_\varepsilon+\psi)/2}.$$

The coefficient curvature is

$$i\Theta_{E^*,k_\varepsilon} = -i\partial\bar\partial(\varphi_\varepsilon + \psi).$$

On $(r, 0)$-forms, the commutator $[-i\partial\bar\partial\varphi_\varepsilon, \Lambda_Y]$ is nonnegative: its eigenvalues are the sums of the complementary $m - r$ eigenvalues of $i\partial\bar\partial\varphi_\varepsilon$. Also $\bar\partial^* v = 0$ by bidegree. The smooth Bochner–Kodaira–Nakano identity therefore gives

$$\|\bar\partial v\|_{k_\varepsilon}^2 \ge \|D'_{k_\varepsilon} v\|_{k_\varepsilon}^2 + \|(D'_{k_\varepsilon})^* v\|_{k_\varepsilon}^2 + \int \langle Qv, v\rangle_{k_\varepsilon}\, dV_Y. \tag{5.2}$$

By the Kähler identity, $(D'_{k_\varepsilon})^*$ is $i[\Lambda_Y, \bar\partial]$, up to the fixed conventional sign. Write this operator as P. It is independent of the coefficient metric, so the second term on the right equals $\|s_\varepsilon P v\|^2$.

To pass to the limit in the connection term, put $t_\varepsilon = e^{\varphi_\varepsilon/2}$. These functions are nonnegative smooth psh functions and are uniformly bounded on compact subballs. For a real compactly supported cutoff η, the Euclidean Laplacian satisfies

$$\Delta(t_\varepsilon^2) = 2t_\varepsilon \Delta t_\varepsilon + 2|\nabla t_\varepsilon|^2 \ge 2|\nabla t_\varepsilon|^2.$$

Integration by parts consequently gives the uniform local bound

$$2\int \eta^2 |\nabla t_\varepsilon|^2 \le \int t_\varepsilon^2 \Delta(\eta^2) \le \big(\sup_{\operatorname{supp}\eta} t_\varepsilon^2\big) \int |\Delta(\eta^2)|.$$

Since $t_\varepsilon \to e^{\varphi/2}$ almost everywhere and the functions are uniformly bounded, dominated convergence and weak compactness give, after passage to a subsequence,

$$s_\varepsilon \longrightarrow s = e^{(\varphi+\psi)/2} \text{ strongly in } L^2_{\mathrm{loc}}, \qquad s_\varepsilon \rightharpoonup s \text{ weakly in } W^{1,2}_{\mathrm{loc}}.$$

More explicitly, on each compact subball K,

$$\int_K |s_\varepsilon - s|^2 \longrightarrow 0, \qquad \sup_\varepsilon \int_K |ds_\varepsilon|^2 < \infty.$$

For every compactly supported smooth test function ζ and every real coordinate x_j, any weak derivative limit a_j satisfies

$$\int a_j \zeta = \lim_\varepsilon \int (\partial_{x_j} s_\varepsilon)\zeta = -\lim_\varepsilon \int s_\varepsilon \partial_{x_j}\zeta = -\int s \partial_{x_j}\zeta.$$

Thus $a_j = \partial_{x_j} s$ distributionally. In the chosen holomorphic frame,

$$D'_{k_\varepsilon} v = \partial v + \partial \log k_\varepsilon \wedge v, \qquad s_\varepsilon D'_{k_\varepsilon} v = s_\varepsilon \partial v + 2\partial s_\varepsilon \wedge v \rightharpoonup T_s v,$$

where $T_s v = s\partial v + 2\partial s \wedge v \in L^2_{\mathrm{loc}}$. The remaining terms of (5.2) converge by dominated convergence: v, Pv, $\bar\partial v$, and Q are smooth on a fixed compact support, and k_ε is uniformly bounded there. In particular,

$$\|T_s v\|^2 \le \liminf_{\varepsilon\to 0} \|s_\varepsilon D'_{k_\varepsilon} v\|^2, \qquad \|s_\varepsilon P v - sPv\| \longrightarrow 0, \qquad \|s_\varepsilon \bar\partial v - s\bar\partial v\| \longrightarrow 0.$$

Passing to the limit in (5.2) therefore yields

$$\|s\bar\partial v\|^2 \ge \|T_s v\|^2 + \|sPv\|^2 + \int \langle Qv, v\rangle_k \, dV_Y. \tag{5.3}$$

The densities in this inequality are invariant under holomorphic frame changes. If the frame of E^* is multiplied by a nowhere vanishing holomorphic function g, then the coefficients and square root of the metric become

$$v' = g^{-1} v, \qquad s' = |g| s.$$

Since $2\partial|g| = (|g|/g)\partial g$, the weak product rule gives

$$T_{s'} v' = \frac{|g|}{g} T_s v, \qquad s' P v' = \frac{|g|}{g} sPv, \qquad s'\bar\partial v' = \frac{|g|}{g} s\bar\partial v.$$

Thus their squared norms agree on overlaps.

Cover $\operatorname{supp} w$ by finitely many such balls and choose real smooth functions ρ_j supported in them with $\sum_j \rho_j^2 = 1$ on a neighborhood of $\operatorname{supp} w$. Apply (5.3) to $\rho_j w$ and sum. The product rules are

$$\bar\partial(\rho_j w) = \rho_j \bar\partial w + \bar\partial \rho_j \wedge w, \qquad T_s(\rho_j w) = \rho_j T_s w + s\partial\rho_j \wedge w,$$

and

$$P(\rho_j w) = \rho_j P w + [P, \rho_j] w, \qquad |[P,\rho_j]w| = |(\partial\rho_j \wedge)^* w|.$$

The last identity follows directly from $P = \pm i[\Lambda_Y, \bar\partial]$; its possible unit factor has no effect on the norm. Every cross term in the sum vanishes because $\sum_j \rho_j \, d\rho_j = 0$; explicitly,

$$\begin{aligned}
\sum_j \|s\bar\partial(\rho_j w)\|^2 &= \|s\bar\partial w\|^2 + \sum_j \|s\bar\partial\rho_j \wedge w\|^2,\\
\sum_j \|T_s(\rho_j w)\|^2 &= \|T_s w\|^2 + \sum_j \|s\partial\rho_j \wedge w\|^2,\\
\sum_j \|sP(\rho_j w)\|^2 &= \|sPw\|^2 + \sum_j \|s[P,\rho_j]w\|^2,\\
\sum_j \int_Y \langle Q(\rho_j w), \rho_j w\rangle_k \, dV_Y &= \int_Y \langle Qw, w\rangle_k \, dV_Y.
\end{aligned}$$

For example, the first cross term is $2\,\mathrm{Re}\int\langle s\bar\partial w, s(\sum_j \rho_j\bar\partial\rho_j)\wedge w\rangle = 0$; the other two have the same linear dependence on $\sum_j \rho_j d\rho_j$. For each real ρ_j, exterior multiplication and contraction satisfy

$$|\bar\partial\rho_j \wedge w|^2 = |\partial\rho_j|^2|w|^2 = |\partial\rho_j \wedge w|^2 + |(\partial\rho_j\wedge)^* w|^2.$$

Consequently the quadratic partition errors on the two sides cancel, leaving

$$\|\bar\partial w\|_k^2 \geq \|T_s w\|^2 + \|sPw\|^2 + \int_Y \langle Qw, w\rangle_k \, dV_Y.$$

The frame covariance just established makes the first two integrals well defined. Discarding them proves (5.1).

Now let f be an E-valued (a,m)-form of finite inverse-B energy. It has locally L^2 coefficients; closedness is automatic in this bidegree. The complementary-index eigenvalue calculation above gives the pointwise wedge-pairing inequality

$$\left|\frac{f\wedge w}{dV_Y}\right| \leq \langle B^{-1}f, f\rangle_h^{1/2}\langle Qw, w\rangle_k^{1/2}.$$

Indeed, the coefficient indexed by I in f pairs only with that indexed by I^c in w, the two curvature factors are β_I^{-1} and β_I, and the line-metric factors are h and $k = h^{-1}$. Integration and Cauchy–Schwarz give

$$\left|\int_Y f\wedge w\right|^2 \leq I_A(f)\int_Y \langle Qw, w\rangle_k \, dV_Y \leq I_A(f)\|\bar\partial w\|_k^2.$$

Define a complex-linear functional on the subspace

$$\mathcal{R} = \{\bar\partial w : w \in C_c^\infty(Y, \Lambda^{r,0}T_Y^*\otimes E^*)\} \subset L^2_{r,1}(Y, E^*, k)$$

by

$$\ell(\bar\partial w) = (-1)^{a+m}\int_Y f\wedge w.$$

The preceding inequality proves that this definition is independent of the choice of w: a difference with $\bar\partial w = 0$ has zero pairing with f. It also gives $\|\ell\| \leq I_A(f)^{1/2}$. Hahn–Banach extends ℓ to the whole Hilbert space with the same bound. Riesz representation, followed by the pointwise isometric wedge duality between $(r,1)$-forms with metric k and $(a, m-1)$-forms with metric $h = k^{-1}$, produces u satisfying

$$\|u\|_h^2 \leq I_A(f), \qquad \int_Y u\wedge\bar\partial w = (-1)^{a+m}\int_Y f\wedge w.$$

The isometry is the usual conjugate Hodge-star identification, combined with the duality $E\otimes E^* \to \mathbb{C}$. Since h is locally bounded below, u is locally L^2 in a smooth metric. Both sides are therefore ordinary distributional pairings. Finally,

$$0 = \int_Y \bar\partial(u\wedge w) = \int_Y \bar\partial u\wedge w + (-1)^{a+m-1}\int_Y u\wedge\bar\partial w$$

is precisely the distributional identity $\bar\partial u = f$. □

Proof of Theorem 1.2. In product coordinate charts, the local weight of h is $\varphi_G(y)+\varphi_M(z)$. The multiplier-ideal product formula [DEL00, Theorem 2.6(i)] gives

$$\mathcal{I}(h) = (\mathrm{pr}_Y^{-1}\mathcal{I}(h_G)\mathcal{O}_X)(\mathrm{pr}_Z^{-1}\mathcal{I}(h_M)\mathcal{O}_X). \tag{5.4}$$

Put

$$\mathcal{A} = G\otimes\mathcal{I}(h_G), \qquad \mathcal{B} = \Omega_Z^p\otimes M\otimes\mathcal{I}(h_M),$$

Then

$$\Omega^p_{X/Y}\otimes L\otimes\mathcal{I}(h) \simeq \mathcal{A}\boxtimes\mathcal{B}.$$

The external tensor product is an ordinary tensor product: the coherent sheaf $\mathrm{pr}_Z^*\,\mathcal{B}$ is flat over Y. Indeed, $\mathcal{B}$ is flat over the one-point base, and its pullback to the product is obtained by base change to Y; flatness is preserved under this base change.

Resolve $\mathcal{A}$ locally on the smooth base by a bounded complex $P^\bullet$ of finite free modules. Such a resolution exists because the local analytic rings of Y are regular. The complex

$$\pi^* P^\bullet \otimes \mathrm{pr}_Z^*\,\mathcal{B}$$

is exact in negative degrees by flatness and resolves $\mathcal{A}\boxtimes\mathcal{B}$. The derived projection formula is therefore applicable to this ordinary external product and gives

$$R\pi_*(\mathcal{A}\boxtimes\mathcal{B}) \simeq \mathcal{A}\otimes^{\mathbf{L}}_{\mathcal{O}_Y} R\pi_*\,\mathrm{pr}_Z^*\,\mathcal{B}.$$

Product base change gives the canonical identifications

$$R^j\pi_*\,\mathrm{pr}_Z^*\,\mathcal{B} \simeq \mathcal{O}_Y\otimes_{\mathbb{C}} H^j(Z,\mathcal{B}).$$

These sheaves are locally free: their fibers are the fixed finite-dimensional cohomology groups of the compact manifold Z. In particular, the projection spectral sequence has terms

$$E_2^{-i,j} = \mathrm{Tor}_i^{\mathcal{O}_Y}\big(\mathcal{A}, \mathcal{O}_Y\otimes_{\mathbb{C}} H^j(Z,\mathcal{B})\big) \Longrightarrow R^{j-i}\pi_*(\mathcal{A}\boxtimes\mathcal{B}),$$

and $E_2^{-i,j}=0$ for $i>0$. It collapses to the column $i=0$, giving

$$R^q\pi_*(\mathcal{A}\boxtimes\mathcal{B}) \simeq \mathcal{A}\otimes_{\mathbb{C}} H^q(Z,\mathcal{B}) = \mathcal{A}\otimes_{\mathbb{C}} V.$$

Together with (5.4), this proves the first equality in (1.2).

Choose a base point outside the analytic cosupport of $\mathcal{I}(h_G)$ at which the weight of h_G is finite. Such points have full measure. On its fiber, the weight differs from the weight of h_M by the finite constant $\varphi_G(y)$, so

$$\mathcal{I}(h|_{X_y}) = \mathcal{I}(h_M), \qquad \|\alpha\|^2_{h|_{X_y},\omega_Z} = h_G(y)\|\alpha\|^2_{h_M,\omega_Z}$$

in a local frame of G. Multiplication of every norm in the fixed fiber complex by the same positive constant leaves its domains and its cohomology unchanged. Taking the minimum over all closed representatives in Lemma 2.6 consequently gives $H_y = h_G(y)H_V$ under the canonical fiber identification.

If $V=0$, the remaining conclusions are immediate. Otherwise choose an H_V-orthonormal basis $e_1,\dots,e_N$. A holomorphic section $s=\sum_{\nu=1}^N s_\nu\otimes e_\nu$ of E_0 has

$$|s|_H^2 = \sum_{\nu=1}^N |s_\nu|^2_{h_G}.$$

Since this is a finite sum of nonnegative functions, its local integrability is equivalent to local integrability of every term. This proves $\mathscr{E}(H) = (G\otimes\mathcal{I}(h_G))\otimes_{\mathbb{C}} V$.

Finally write a datum in either stated extremal degree as $f=\sum_{\nu=1}^N f_\nu\otimes e_\nu$. The basis is constant and holomorphic, hence $\bar\partial f=0$ implies $\bar\partial f_\nu=0$ for every ν. The auxiliary curvature acts trivially on V, so

$$B_A = B_{A,G}\otimes \mathrm{Id}_V, \qquad I_A(f) = \sum_{\nu=1}^N I_A(f_\nu) < \infty.$$

Lemma 5.1 supplies solutions $\bar\partial u_\nu = f_\nu$ with $\|u_\nu\|^2_{h_G h_A} \le I_A(f_\nu)$. For $u=\sum_\nu u_\nu\otimes e_\nu$, orthogonality gives

$$\bar\partial u = f, \qquad \|u\|^2_{Hh_A} = \sum_{\nu=1}^N \|u_\nu\|^2_{h_G h_A} \le \sum_{\nu=1}^N I_A(f_\nu) = I_A(f).$$

This proves both families of estimates for every p,q. Proposition 8.1, applied to $H=h_G\otimes H_V$ and $\mathcal{F}=(G\otimes\mathcal{I}(h_G))\otimes V$, proves Griffiths semipositivity and the minimal extension property. □

Remark 5.2. The fiberwise norm identification holds almost everywhere. At a base point where $h_G = +\infty$, the restricted weight is identically $-\infty$ on the fiber. The coherent sheaf in (1.2) is defined on all of Y.

6. Top adjoint descent for arbitrary singular semipositive metrics

Let $\pi : X \to Y$ be a proper surjective holomorphic submersion with connected fibers of dimension $n \geq 1$, where X and Y are connected complex manifolds. Fix a Kähler form ω_X on X and put $\omega_y = \omega_X|_{X_y}$. For a singular Hermitian line bundle (L, h) with psh local weights, write

$$T_h = \frac{i}{2\pi}\Theta_{L,h}, \qquad \mathcal{F}_n = R^n\pi_*(K_{X/Y} \otimes L \otimes \mathcal{I}(h)).$$

Thus $dd^c = (i/2\pi)\partial\bar{\partial}$ in this section. This normalization gives $T_{h_D} = [D]$ for the canonical singular metric h_D on $\mathcal{O}_X(D)$: if $s_D = fe_D$ is the canonical section, then $|e_D|^2_{h_D} = |f|^{-2}$ and $|s_D|_{h_D} = 1$ off D.

The proof of Theorem 1.3 uses the following two sheaf-theoretic facts.

Lemma 6.1 (Top-degree duality and projection)**.** *For a coherent sheaf $\mathcal{A}$ on X,*

$$\mathcal{H}om_Y\Big(R^n\pi_*(K_{X/Y} \otimes \mathcal{A}), \mathcal{O}_Y\Big) \simeq \pi_*\mathcal{H}om_X(\mathcal{A}, \mathcal{O}_X). \tag{6.1}$$

For a coherent sheaf $\mathcal{B}$ on Y,

$$R^n\pi_*(K_{X/Y} \otimes \pi^*\mathcal{B}) \simeq \mathcal{B}. \tag{6.2}$$

Proof. Put $C = R\pi_*(K_{X/Y} \otimes \mathcal{A})$, with cohomological indexing, so that $\mathcal{H}^k(C) = 0$ unless $0 \leq k \leq n$; the upper bound is the coherent cohomological-dimension theorem for a proper map with n-dimensional fibers. Work locally on the smooth base. Coherent sheaves on X and Y have finite locally free resolutions locally. The smooth morphism π has relative dualizing complex $\pi^!\mathcal{O}_Y = K_{X/Y}[n]$. Analytic relative duality [RRV71] therefore gives

$$\begin{aligned} R\mathcal{H}om_Y(C, \mathcal{O}_Y) &\simeq R\pi_*R\mathcal{H}om_X(K_{X/Y} \otimes \mathcal{A}, K_{X/Y}[n]) \\ &\simeq R\pi_*R\mathcal{H}om_X(\mathcal{A}, \mathcal{O}_X)[n]. \end{aligned}$$

The invertible factor $K_{X/Y}$ cancels inside the derived Hom. Our shift convention is $\mathcal{H}^j(Q[n]) = \mathcal{H}^{j+n}(Q)$. On the left the spectral sequence is

$$\begin{gathered} E_2^{p,q} = \mathcal{E}xt^p_Y(\mathcal{H}^{-q}(C), \mathcal{O}_Y) \Longrightarrow \mathcal{H}^{p+q}(R\mathcal{H}om_Y(C, \mathcal{O}_Y)), \\ p \geq 0, \qquad -n \leq q \leq 0. \end{gathered}$$

Total degree $-n$ has only $(p, q) = (0, -n)$. An incoming differential to this term has negative first index, whereas an outgoing differential d_r has second index $-n - r + 1 < -n$. Thus the degree $-n$ edge map identifies its abutment with $\mathcal{H}om_Y(R^n\pi_*(K_{X/Y} \otimes \mathcal{A}), \mathcal{O}_Y)$. On the right, before shifting, the spectral sequence

$${}'E_2^{a,b} = R^a\pi_*\mathcal{E}xt^b_X(\mathcal{A}, \mathcal{O}_X) \Longrightarrow \mathcal{H}^{a+b}(R\pi_*R\mathcal{H}om_X(\mathcal{A}, \mathcal{O}_X))$$

has $a, b \geq 0$. At $(0, 0)$ the source of an incoming d_r has first index $-r$, and the target of an outgoing d_r has second index $1 - r$; both vanish for $r \geq 2$. Its degree-zero edge term is therefore $\pi_*\mathcal{H}om_X(\mathcal{A}, \mathcal{O}_X)$. Taking degree $-n$ after the shift proves (6.1).

We record the map underlying this isomorphism. Normalize the relative trace

$$\mathrm{Tr}_\pi : R^n\pi_*K_{X/Y} \longrightarrow \mathcal{O}_Y$$

so that its fiber map is integration of Dolbeault (n, n)-classes. If $\lambda : \mathcal{A}|_{\pi^{-1}(U)} \to \mathcal{O}_{\pi^{-1}(U)}$ is a sheaf morphism, the duality map sends it to

$$\mathrm{Tr}_\pi \circ R^n\pi_*(\mathrm{Id}_{K_{X/Y}} \otimes \lambda) : R^n\pi_*(K_{X/Y} \otimes \mathcal{A})|_U \longrightarrow \mathcal{O}_U. \tag{6.3}$$

This is the evaluation pairing followed by the trace in relative duality. Its functoriality fixes the cohomological identifications used below.

The submersion π is flat. Since Y is smooth, $\mathcal{B}$ also has local finite locally free resolutions, so the derived projection formula gives

$$R\pi_*(K_{X/Y}\otimes\pi^*\mathcal{B})\simeq(R\pi_*K_{X/Y})\otimes^{\mathbf{L}}\mathcal{B}.$$

Here ordinary $\pi^*\mathcal{B}$ equals its derived pullback. The tensor spectral sequence has terms

$$''E_2^{-j,k}=\mathcal{T}or_j^{\mathcal{O}_Y}(R^k\pi_*K_{X/Y},\mathcal{B}),\qquad j\geq 0,\quad 0\leq k\leq n,$$

and abuts in degree $k-j$. In degree n its only term is $''E_2^{0,n}=(R^n\pi_*K_{X/Y})\otimes\mathcal{B}$. An outgoing d_r has positive first index; an incoming d_r starts at $(-r,n+r-1)$, whose second index exceeds n. Both vanish. Fiberwise Serre duality gives $h^n(X_y,K_{X_y})=h^0(X_y,\mathcal{O}_{X_y})=1$ for every y. Grauert's theorem therefore makes $R^n\pi_*K_{X/Y}$ a line bundle with base change. Its trace map is an isomorphism on every fiber, hence

$$R^n\pi_*K_{X/Y}\simeq\mathcal{O}_Y.$$

The preceding degree-n edge isomorphism followed by $\mathrm{Tr}_\pi\otimes\mathrm{Id}_{\mathcal{B}}$ proves (6.2). In particular, if $\mathcal{B}$ is a line bundle near y, its fiber map is

$$H^n(X_y,K_{X_y})\otimes\mathcal{B}_y\longrightarrow\mathcal{B}_y,\qquad[v]\otimes e\longmapsto\left(\int_{X_y}v\right)e.$$

□

Proof of Theorem 1.3. On the smooth space X, the double dual of the coherent rank-one ideal $\mathcal{I}(h)$ is an invertible ideal $\mathcal{O}_X(-D)$, where D is effective. Factoring out this invertible ideal gives

$$\mathcal{I}(h)=\mathcal{O}_X(-D)\mathcal{J},\qquad\operatorname{codim}_X\operatorname{Supp}(\mathcal{O}_X/\mathcal{J})\geq 2.\tag{6.4}$$

For a prime divisor P, its coefficient in D is $\lfloor\nu_P(T_h)\rfloor$. To verify this for arbitrary psh weights, put $a=\nu_P(T_h)$. Near a smooth generic point of $P=\{z_1=0\}$, Siu decomposition gives

$$\varphi=a\log|z_1|^2+\eta,\qquad\eta\text{ psh},\qquad\nu_P(dd^c\eta)=0.$$

For $k=\lfloor a\rfloor$, put

$$u_k=\tfrac{1}{2}(\varphi-k\log|z_1|^2).$$

In the convention $d^c=\frac{i}{2\pi}(\bar\partial-\partial)$ used in [Dem12], the psh function u_k has Lelong number $a-k<1$ at generic points of P. Skoda's criterion [Dem12, Lemma 5.6(a)] therefore gives

$$e^{-2u_k}=|z_1|^{2k}e^{-\varphi}\in L^1_{\mathrm{loc}},\qquad\operatorname{ord}_P\mathcal{I}(h)\leq k$$

there. The factor $1/2$ converts the weight convention in that criterion to the convention $h=e^{-\varphi}$ used here. Conversely, if $g=z_1^\ell u$ with u a unit and $\ell<k$, local upper bounds for η and a lower bound for $|u|$ give

$$|g|^2e^{-\varphi}\geq c|z_1|^{-2(a-\ell)},\qquad\int_0^\epsilon r^{1-2(a-\ell)}\,dr=+\infty,\quad a-\ell\geq 1.$$

Thus $g\notin\mathcal{I}(h)$ and the generic order is exactly k. Coherence makes D locally finite. Subtracting these integer parts from the divisorial terms in Siu decomposition gives

$$M=L(-D),\qquad h_M=h/h_D,\qquad T_{h_M}=T_h-[D]\geq 0.\tag{6.5}$$

The residual weights, initially defined almost everywhere, have psh representatives because their distributional complex Hessians are positive. We use these representatives from now on. There is also an equality of multiplier ideals

$$\mathcal{J}=\mathcal{I}(h_M),\qquad\mathcal{I}(h)=\mathcal{O}_X(-D)\mathcal{I}(h_M).$$

Indeed, write $s_D = fe_D$ and $\varphi_h = \log|f|^2 + \varphi_M$ in compatible frames. Then

$$|g|^2 e^{-\varphi_h} = |g/f|^2 e^{-\varphi_M}.$$

If this is integrable, the local upper bound for the psh function φ_M implies $g/f \in L^2_{\mathrm{loc}}$ off D. A meromorphic function with this property has no divisorial poles: in a transverse coordinate a pole of order $r \geq 1$ has divergent integral $\int_0^\epsilon t^{1-2r}\,dt$. It then extends across the remaining codimension-two set by Hartogs. Thus g/f is holomorphic and belongs to $\mathcal{I}(\varphi_M)$. The reverse implication follows from the same displayed equality.

Matsumura's torsion-freeness theorem [Mat22, Corollary 1.5] applies because the restrictions of ω_X make π a proper Kähler morphism in the sense of [Mat22, Remark 1.4(1)]. The projection formula and $K_X = K_{X/Y} \otimes \pi^* K_Y$ give

$$\mathcal{F}_n \simeq R^n\pi_*(K_X \otimes L \otimes \mathcal{I}(h)) \otimes K_Y^{-1},$$

so $\mathcal{F}_n$ is torsion-free. Suppose $\mathcal{F}_n \neq 0$. Its generic rank is positive. To obtain a nonzero dual germ at an arbitrary $y \in Y$, put $R = \mathcal{O}_{Y,y}$ and $K = \mathrm{Frac}(R)$. Choose a nonzero K-linear functional on $(\mathcal{F}_n)_y \otimes_R K$. If $v_1, \ldots, v_N$ generate $(\mathcal{F}_n)_y$ and $\ell(v_j) = a_j/b_j$, put $d = \prod_j b_j \neq 0$. Then

$$0 \neq d\ell \in \mathrm{Hom}_R((\mathcal{F}_n)_y, R).$$

Coherence extends this germ to a section of $\mathcal{F}_n^*$ near y. Moreover,

$$\mathcal{H}om_X(\mathcal{J}, \mathcal{O}_X) = \mathcal{O}_X.$$

In fact, away from the codimension-two cosupport of $\mathcal{J}$, such a homomorphism is multiplication by a holomorphic function. Hartogs extends that function to X, and the extended multiplication agrees with the homomorphism because $\mathcal{O}_X$ is torsion-free. Applying (6.1) to $\mathcal{A} = M \otimes \mathcal{J}$ therefore yields

$$\mathcal{F}_n^* \simeq \pi_* M^{-1}. \tag{6.6}$$

After shrinking to a connected neighborhood U of y, we obtain a nonzero $\sigma \in H^0(\pi^{-1}(U), M^{-1})$. The space $\pi^{-1}(U)$ is connected: the map is proper and open, and its fibers are connected. Thus σ defines the effective divisor $E = \mathrm{div}(\sigma)$.

We next show that every component of E is vertical. Cover $\pi^{-1}(U)$ by countably many relatively compact submersion charts $V_\alpha \times W_\alpha$, in which M is trivial and the psh weights φ_α are integrable. Fubini gives measure-zero sets $N_\alpha \subset V_\alpha$ such that

$$\begin{aligned} &\int_{V_\alpha}\int_{W_\alpha} |\varphi_\alpha(y,z)|\,dV_z\,dV_y < \infty \\ \Longrightarrow\quad &\int_{W_\alpha} |\varphi_\alpha(y,z)|\,dV_z < \infty \quad (y \notin N_\alpha). \end{aligned}$$

Using a countable exhaustion in each chart, put $U_0 = U \setminus \bigcup_\alpha N_\alpha$. For $y \in U_0$ all restricted weights are locally integrable psh functions and define a semipositive metric on M_y. Also put $S = \{y \in U : \dim(E \cap X_y) = n\}$. Properness and the fiber-dimension theorem make S a proper analytic subset. For $y \in U_0 \setminus S$, the section σ_y is nonzero and has effective divisor E_y. Consequently

$$0 \leq \int_{X_y} T_{h_M|_{X_y}} \wedge \omega_y^{n-1} = \int_{X_y} c_1(M_y) \wedge \omega_y^{n-1} = -\int_{E_y} \omega_y^{n-1} \leq 0. \tag{6.7}$$

For the middle equality, choose a smooth reference metric h_0 on M_y and write $h_M|_{X_y} = h_0 e^{-\chi}$, with $\chi \in L^1(X_y)$. Since $d\omega_y = 0$ and X_y is compact,

$$\int_{X_y} dd^c\chi \wedge \omega_y^{n-1} = \langle \chi, dd^c\omega_y^{n-1}\rangle = 0.$$

The next equality uses $M_y^{-1} \simeq \mathcal{O}_{X_y}(E_y)$. If a component of E dominated U, its generic fiber would be a nonzero effective divisor. Positivity of its Kähler volume would contradict (6.7). Thus E is vertical.

Let E_0 be a vertical prime component and $B_0 = \pi(E_0)$. By properness, B_0 is irreducible analytic, and the fiber-dimension bound gives

$$\dim B_0 \geq \dim E_0 - n = \dim U - 1.$$

Hence B_0 is a prime divisor. Over its connected smooth locus, $\pi^{-1}(B_0)$ is a connected complex manifold: the restricted map is a proper submersion with connected fibers. It is therefore irreducible. The closed analytic subset E_0 has the same dimension and dominates B_0, so it equals the closure of this inverse image. If b_0 locally defines a smooth point of B_0, the submersion property gives $d(b_0 \circ \pi) \neq 0$ there. Thus the pullback divisor is generically reduced; no additional divisorial component can lie over the codimension-two singular locus of B_0. Assigning to B_0 the multiplicity of E_0 in E proves

$$E = \pi^* B \tag{6.8}$$

for an effective divisor B on U. Properness makes these image components locally finite: a compact inverse image of a compact base neighborhood meets only finitely many local components of E. Shrink U so that $B = (b)$. The meromorphic section $\tau_U = \sigma/(b \circ \pi)$ has order zero along every prime divisor. In a local frame, its coefficient and the reciprocal coefficient are holomorphic off a codimension-two set and extend by Hartogs. Thus τ_U is a nowhere-vanishing holomorphic frame of M^{-1}. In the dual frame $e_U = \tau_U^{-1}$, write $|e_U|^2_{h_M} = e^{-\varphi_U}$. The weight φ_U is a global psh function on $\pi^{-1}(U)$.

Its restriction to each compact connected fiber is constant or identically $-\infty$. Define $\psi_U(y)$ to be that value; then

$$\varphi_U = \psi_U \circ \pi. \tag{6.9}$$

For a local holomorphic section $s : W \to \pi^{-1}(U)$ of the submersion, $\psi_U|_W = \varphi_U \circ s$ is psh. It cannot be identically $-\infty$ on an open W, since that would make φ_U identically $-\infty$ on the open set $\pi^{-1}(W)$. On overlaps write $e_V = a_{UV} e_U$. The holomorphic unit a_{UV} is constant on each compact connected fiber, so $a_{UV} = g_{UV} \circ \pi$ for a holomorphic unit g_{UV} on $U \cap V$. This also follows from $\pi_* \mathcal{O}_X = \mathcal{O}_Y$. The frame identities and metric identities give, respectively,

$$g_{UV} g_{VW} = g_{UW}, \qquad \psi_V = \psi_U - \log |g_{UV}|^2.$$

Thus the g_{UV} define a line bundle G, with frames $\epsilon_V = g_{UV} \epsilon_U$, and $|\epsilon_U|^2_{h_G} = e^{-\psi_U}$ defines a psh metric on it. The maps $\pi^* \epsilon_U \mapsto e_U$ glue to

$$(M, h_M) \simeq \pi^*(G, h_G),$$

which proves (1.3).

The residual multiplier-ideal identity proved above now reads

$$\mathcal{I}(h) = \mathcal{O}_X(-D) \mathcal{I}(\psi_U \circ \pi).$$

In submersion coordinates (y, z) the weight is $\psi_U(y) + 0(z)$. The multiplier-ideal product formula [DEL00, Theorem 2.6(i)] therefore gives

$$\mathcal{I}(\psi_U \circ \pi) = \pi^{-1} \mathcal{I}(\psi_U) \mathcal{O}_X.$$

These equalities glue, proving the first part of (1.4). In particular there is an isomorphism of the actual coherent sheaves

$$K_{X/Y} \otimes L \otimes \mathcal{I}(h) \simeq K_{X/Y} \otimes \pi^*(G \otimes \mathcal{I}(h_G)).$$

Apply (6.2) to $\mathcal{B} = G \otimes \mathcal{I}(h_G)$. This proves $\mathcal{F}_n \simeq G \otimes \mathcal{I}(h_G)$.

We identify this sheaf isomorphism and its fiber maps before computing the norm. Put

$$\mathcal{B} = G \otimes \mathcal{I}(h_G), \qquad \mathcal{C} = K_{X/Y} \otimes L \otimes \mathcal{I}(h).$$

Under (1.3), multiplication by s_D is the holomorphic isomorphism

$$\mu : K_{X/Y} \otimes \pi^*\mathcal{B} \xrightarrow{\sim} \mathcal{C}, \qquad \eta \otimes g \longmapsto s_D\eta \otimes g. \tag{6.10}$$

Locally it sends a coefficient a to fa, where $s_D = fe_D$; the identity $\mathcal{I}(h) = f\,\pi^*\mathcal{I}(h_G)$ proves both surjectivity and injectivity. Fix the isomorphism

$$\Phi : \mathcal{F}_n \xrightarrow{(R^n\pi_*\mu)^{-1}} R^n\pi_*(K_{X/Y} \otimes \pi^*\mathcal{B}) \xrightarrow{\mathrm{Tr}_\pi} \mathcal{B},$$

where the second arrow is the projection isomorphism of Lemma 6.1, normalized by its trace.

Define the exceptional sets

$$\begin{aligned} S_I &= \mathrm{Supp}(\mathcal{O}_Y/\mathcal{I}(h_G)), \qquad P_G = \{y : \psi_U(y) = -\infty\}, \\ S_D &= \{y : \dim(|D| \cap X_y) = n\} = \{y : X_y \subset |D|\}, \\ Y_{\mathrm{good}} &= Y \setminus (S_I \cup S_D \cup P_G). \end{aligned}$$

The definition of P_G is independent of the local frame. The set S_I is proper analytic and P_G has measure zero. Properness of $D \to Y$ and the fiber-dimension theorem make S_D analytic. It is proper, since $S_D = Y$ would imply $|D| = X$. Connectedness of X_y justifies the second expression for S_D. Consequently Y_{good} has full measure. This choice also excludes whole fibers contained in horizontal components of D.

For $y \in Y_{\mathrm{good}}$, the restriction $D_y = D|_{X_y}$ is an effective Cartier divisor, and $h_G(y)$ is positive and finite. In compatible local frames, the factorization gives

$$\varphi_h = \log|f_j|^2 + \psi_U \circ \pi$$

as an equality of psh functions everywhere: both sides are psh and agree almost everywhere. Its restriction to each such fiber gives

$$(L_y, h_y) = (\mathcal{O}_{X_y}(D_y), h_{D_y}) \otimes (G_y, h_G(y)).$$

The preceding divisor integrability argument, applied on X_y, shows that

$$\mathcal{I}(h_y) = \mathcal{O}_{X_y}(-D_y) = \mathcal{I}(h)\mathcal{O}_{X_y}. \tag{6.11}$$

In the second equality we use $y \notin S_I$ and that a local equation of D restricts to a nonzero divisor on the smooth connected fiber. In particular, writing $i_y : X_y \hookrightarrow X$, the natural map

$$i_y^*\mathcal{C} \longrightarrow K_{X_y} \otimes L_y$$

is locally multiplication by the nonzero divisor $f_j|_{X_y}$. It is injective, with image $K_{X_y} \otimes L_y \otimes \mathcal{I}(h_y)$. Thus the restriction of (6.10) is

$$\mu_y : K_{X_y} \otimes G_y \xrightarrow{\sim} K_{X_y} \otimes L_y \otimes \mathcal{I}(h_y).$$

Set $Y' = Y \setminus S_I$. On $\pi^{-1}(Y')$, the source of μ is $K_{X/Y} \otimes \pi^*G$. Naturality of the cohomological base-change maps gives, for $y \in Y_{\mathrm{good}}$, the commutative diagram

$$\begin{array}{ccc} (R^n\pi_*(K_{X/Y} \otimes \pi^*G)) \otimes \mathbb{C}(y) & \xrightarrow{(R^n\pi_*\mu)_y} & \mathcal{F}_n \otimes \mathbb{C}(y) \\ \downarrow \beta_y^0 & & \downarrow \beta_y^h \\ H^n(X_y, K_{X_y}) \otimes G_y & \xrightarrow{H^n(\mu_y)} & H^n(X_y, K_{X_y} \otimes L_y \otimes \mathcal{I}(h_y)). \end{array}$$

Both horizontal arrows are isomorphisms. The map β_y^0 is an isomorphism by Grauert's theorem, since $h^n(X_y, K_{X_y}) = 1$. Hence β_y^h is an isomorphism as well. This proves that the fiber space carrying the prescribed norm is identified with the fiber of the actual direct image by its canonical

base-change map. Under this identification, put $\widehat{\Phi}_y = \Phi_y \circ (\beta_y^h)^{-1}$. Compatibility of the relative trace with the fiber trace gives

$$\widehat{\Phi}_y(H^n(\mu_y)([v]\otimes\epsilon)) = \left(\int_{X_y} v\right)\epsilon. \tag{6.12}$$

Fix $y \in Y_{\text{good}}$ and $0 \neq \epsilon \in G_y$, and write $c_y = |\epsilon|^2_{h_G(y)}$. Let

$$\mathcal{H}_r = L^2_{n,r}(X_y, \omega_y), \qquad \mathcal{K}_r = L^2_{n,r}(X_y, L_y; h_y, \omega_y), \qquad 0 \le r \le n.$$

These spaces consist of equivalence classes modulo sets of measure zero. If $s_D = f_j e_j$ and $u = u_j \otimes (e_j \otimes \epsilon)$ locally, division by f_j on $X_y \setminus D_y$ defines compatible scalar forms u_j/f_j. Multiplication by $s_D \otimes \epsilon$ is therefore a bijection of Hilbert spaces

$$T_r : \mathcal{H}_r \xrightarrow{\sim} \mathcal{K}_r, \qquad T_r v = s_D v \otimes \epsilon, \qquad \|T_r v\|^2_{h_y} = c_y \|v\|^2. \tag{6.13}$$

Indeed the pointwise equality off D_y is

$$|f_j v|^2 \, |f_j|^{-2} c_y = c_y |v|^2,$$

and D_y has measure zero; the inverse map is division by the local equations followed by any measurable extension on D_y.

Let d_r and d_r^h be the maximal distributional $\bar\partial$ operators on $\mathcal{H}_r$ and $\mathcal{K}_r$, respectively. Set $\mathcal{H}_{n+1} = \mathcal{K}_{n+1} = 0$ and $T_{n+1} = 0$. We verify the equality of graph domains. For $v \in \operatorname{Dom} d_r$, the distributional product rule with the holomorphic function f_j gives

$$\bar\partial(f_j v) = f_j \bar\partial v, \qquad T_r v \in \operatorname{Dom} d_r^h, \qquad d_r^h T_r v = T_{r+1} d_r v.$$

Conversely, take $u \in \operatorname{Dom} d_r^h$ and set $v = T_r^{-1} u$ and $w = T_{r+1}^{-1} d_r^h u$. By (6.13), v, w are ordinary L^2 forms, and off D_y,

$$\bar\partial v = \bar\partial(u_j/f_j) = (\bar\partial u_j)/f_j = w.$$

Lemma 2.5 gives $\bar\partial v = w$ on the entire fiber. Thus

$$\begin{aligned} &T_r(\operatorname{Dom} d_r) = \operatorname{Dom} d_r^h, \qquad d_r^h T_r = T_{r+1} d_r, \\ &\|T_r v\|^2_{h_y} + \|d_r^h T_r v\|^2_{h_y} = c_y(\|v\|^2 + \|d_r v\|^2). \end{aligned} \tag{6.14}$$

In particular $U_r = c_y^{-1/2} T_r$ is unitary, and taking Hilbert adjoints in $d_r^h U_r = U_{r+1} d_r$ gives, with equality of domains,

$$(d_r^h)^* U_{r+1} = U_r d_r^*.$$

It follows that T_r takes kernels, images, and harmonic subspaces onto their weighted counterparts. The ordinary compact Dolbeault complex has closed range; hence so does the weighted complex.

The induced cohomology map is exactly $H^n(\mu_y)$ in the preceding base-change diagram. To see this, use the fine resolutions by locally L^2 maximal graph sheaves in Lemma 2.6. The same local calculation proves that $T_\bullet$ is a morphism of these resolutions. On their degree-zero kernels its map is

$$K_{X_y} \otimes G_y \longrightarrow K_{X_y} \otimes L_y \otimes \mathcal{I}(h_y), \qquad \eta \otimes \epsilon \longmapsto s_D \eta \otimes \epsilon,$$

which is μ_y. Functoriality of the resolution isomorphism gives

$$[T_n v] = H^n(\mu_y)([v] \otimes \epsilon).$$

Thus the weighted complex realizes the cohomological identification in (6.12).

Put $dV_y = \omega_y^n/n!$ and $V_y = \int_{X_y} dV_y$. The ordinary maximal Dolbeault quotient is $H^n(X_y, K_{X_y})$, and its trace is $[v] \mapsto \int_{X_y} v$. For $a \in \operatorname{Dom} d_{n-1}$, distributional Stokes on the compact fiber gives

$$\int_{X_y} d_{n-1} a = 0.$$

Serre duality with $H^0(X_y, \mathcal{O}_{X_y}) = \mathbb{C}$ makes this trace an isomorphism. Thus the representatives of a fixed class of trace λ are precisely the L^2 top forms with integral λ. For a unitary coframe $\zeta^1, \dots, \zeta^n$,

$$dV_y = i^{n^2} \zeta^1 \wedge \cdots \wedge \zeta^n \wedge \overline{\zeta^1} \wedge \cdots \wedge \overline{\zeta^n}, \qquad |dV_y|^2 = 1.$$

Consequently every $v = a\, dV_y$ satisfies

$$\left| \int_{X_y} v \right|^2 = \left| \int_{X_y} a\, dV_y \right|^2 \le V_y \int_{X_y} |a|^2\, dV_y = V_y \|v\|^2,$$

with equality for $v = (\lambda/V_y) dV_y$ when its trace is λ. Combining this calculation with the bijection of graph domains and the cohomological identification yields the exact minimization formula

$$\begin{aligned} |\lambda\epsilon|^2_{H_y} &= \min_{\widehat{\Phi}_y([u]) = \lambda\epsilon} \|u\|^2_{h_y} \\ &= c_y \min_{\int_{X_y} v = \lambda} \|v\|^2 = \frac{c_y |\lambda|^2}{V_y}. \end{aligned} \tag{6.15}$$

The minimizing weighted representative is $s_D(\lambda/V_y) dV_y \otimes \epsilon$. This expression is independent of the chosen frame: replacing ϵ by $a\epsilon$ replaces c_y by $|a|^2 c_y$ and λ by $a^{-1}\lambda$.

Finally, fiber integration under the proper submersion gives a smooth function $V_y = \pi_*(\omega_X^n/n!)$, and

$$d\, \pi_*(\omega_X^n/n!) = \pi_* d(\omega_X^n/n!) = 0.$$

Connectedness of Y makes it a constant $V > 0$. Equation (6.15) proves $H = V^{-1} h_G$ on Y_{good}, and hence almost everywhere, as asserted in (1.5). In a frame of G, local square integrability for this metric is exactly $|g|^2 e^{-\psi} \in L^1_{\mathrm{loc}}$; the constant V^{-1} does not affect it. Therefore $\mathscr{E}(H) = G \otimes \mathcal{I}(h_G) = \mathcal{F}_n$.

Finally, for a positive auxiliary line A and a closed (m, b) datum of finite inverse energy, Lemma 5.1 gives $\bar\partial u = f$ and

$$\|u\|^2_{Hh_A} = V^{-1} \|u\|^2_{h_G h_A} \le V^{-1} \int_Y \langle B_A^{-1} f, f \rangle_{h_G h_A}\, dV_Y = \int_Y \langle B_A^{-1} f, f \rangle_{H h_A}\, dV_Y.$$

This is the prescribed weighted optimal estimate. □

Remark 6.2. For disconnected fibers the argument applies locally to their connected components, and the resulting metrics form an orthogonal direct sum.

7. Counterexamples to unrestricted descent

Proposition 7.1. *There is a complete Kähler product $X = Y \times \mathbb{P}^1$ and a smooth metric on $L = \mathrm{pr}_{\mathbb{P}^1}^* \mathcal{O}(2)$ such that*

$$(L, h) \text{ is optimal } (3, 2), \qquad (\pi_*(K_{X/Y} \otimes L), H) \text{ is not optimal } (2, 2).$$

This is range (i) with $m = 2, n = p = 1, a = b = 2, q = 0$.

Proof. Put $r = 1/48$, $Y = \Delta_r \times \Delta$, and

$$\omega_Y = \frac{i\, dt \wedge d\bar t}{(1 - |t|^2/r^2)^2} + \frac{i\, ds \wedge d\bar s}{(1 - |s|^2)^2}, \qquad \psi_Y = -r^2 \log(1 - |t|^2/r^2) - \log(1 - |s|^2). \tag{7.1}$$

Differentiation gives

$$\partial_t \bar\partial_t [-r^2 \log(1 - |t|^2/r^2)] = (1 - |t|^2/r^2)^{-2}.$$

The analogous identity holds in s, so $\omega_Y = i\partial\bar\partial\psi_Y$. Each disk metric is complete: the length of a curve approaching its boundary dominates the radial integral $\int_0^R (1-\rho^2/R^2)^{-1}d\rho = +\infty$. The product metric is therefore complete. On $Z = \mathbb{P}^1$ set

$$\omega_Z = \frac{i\,dz\wedge d\bar z}{(1+|z|^2)^2}, \quad \chi = \frac{1-|z|^2}{1+|z|^2}, \quad h_0 = (1+|z|^2)^{-2} \text{ on } M = K_Z^{-1}.$$

The expression for χ in the coordinate $\zeta = 1/z$ is $(|\zeta|^2-1)/(|\zeta|^2+1)$, so it extends smoothly over infinity. Its coordinate derivatives are

$$\chi_z = -\frac{2\bar z}{(1+|z|^2)^2}, \qquad \chi_{z\bar z} = -\frac{2(1-|z|^2)}{(1+|z|^2)^3}.$$

Consequently

$$|\partial\chi|^2_{\omega_Z} = \frac{4|z|^2}{(1+|z|^2)^2} \le 1, \qquad i\partial\bar\partial\chi = -2\chi\omega_Z. \tag{7.2}$$

Use $\omega_X = \omega_Y + \omega_Z$ and

$$h = h_0 e^{-6(t+\bar t)\chi}(1-|s|^2)^6. \tag{7.3}$$

Both X and Y admit positive auxiliary lines: use $e^{-\psi_Y}$ on the base and its external product with $\mathcal{O}_Z(1)$ on X.

The weight of h in the standard $\mathcal{O}(2)$ frame is

$$\varphi = 2\log(1+|z|^2) + 6(t+\bar t)\chi - 6\log(1-|s|^2).$$

Use the unit tangent vectors

$$e_t = (1-|t|^2/r^2)\partial_t, \quad e_z = (1+|z|^2)\partial_z, \quad e_s = (1-|s|^2)\partial_s.$$

The only mixed coefficient is

$$B = \varphi_{t\bar z}(1-|t|^2/r^2)(1+|z|^2) = -\frac{12z(1-|t|^2/r^2)}{1+|z|^2}, \qquad |B| \le 6.$$

The diagonal coefficients are 0, $2-12(t+\bar t)\chi$, and 6. Thus the curvature matrix is

$$\begin{pmatrix} 0 & B & 0 \\ \bar B & c & 0 \\ 0 & 0 & 6 \end{pmatrix}, \qquad |B| \le 6, \quad c = 2-12(t+\bar t)\chi \in (3/2, 5/2). \tag{7.4}$$

If $\lambda_- \le \lambda_+$ are the first block's eigenvalues, $\lambda_- + \lambda_+ = c > 0$; here $|12(t+\bar t)\chi| < 24r = 1/2$ gives the asserted interval for c. Moreover,

$$\det\begin{pmatrix} 6 & B \\ \bar B & 6+c \end{pmatrix} = 36 + 6c - |B|^2 \ge 6c > 0,$$

The first principal minor of this shifted block is $6 > 0$, so the positive determinant makes the block positive definite. Its eigenvalues are $6+\lambda_-$ and $6+\lambda_+$, whence $\lambda_- > -6$. All three pair sums

$$c, \qquad 6+\lambda_-, \qquad 6+\lambda_+ \tag{7.5}$$

are positive. Equations (2.1) and Proposition 2.2 prove the upstairs estimate.

The holomorphic section $dz\otimes\partial_z$ is the global identity section of $K_Z\otimes K_Z^{-1}$ and generates its one-dimensional space of sections. It has norm one for (ω_Z, h_0), because

$$|dz|^2_{\omega_Z}\,|\partial_z|^2_{h_0} = (1+|z|^2)^2(1+|z|^2)^{-2} = 1.$$

In cohomological degree zero the representative is unique, so the prescribed direct-image metric is its integral:

$$H(t,s) = (1-|s|^2)^6 \int_Z e^{-6(t+\bar t)\chi}\,\omega_Z.$$

With $z = \rho e^{i\theta}$, $\omega_Z = 2\rho(1+\rho^2)^{-2} d\rho\, d\theta$ and $d\chi = -4\rho(1+\rho^2)^{-2} d\rho$. Therefore the pushforward of $\omega_Z/(2\pi)$ under χ is $dx/2$ on $[-1,1]$, giving

$$H(t,s) = 2\pi(1-|s|^2)^6 \frac{\sinh u}{u}, \qquad u = 6(t+\bar{t}). \tag{7.6}$$

Put $F(u) = \log(\sinh u/u)$, with its smooth value at zero. Then $F''(0) = 1/3$ and

$$-\partial_t\bar{\partial}_t \log H(0) = -36F''(0) = -12, \qquad -\partial_s\bar{\partial}_s \log H(0) = 6, \qquad \partial_t\bar{\partial}_s \log H = 0.$$

Since $\omega_Y(0)$ is the identity,

$$i\Theta_H(0) = -12i\, dt \wedge d\bar{t} + 6i\, ds \wedge d\bar{s}, \qquad [i\Theta_H, \Lambda_Y]_{2,2}(0) = -6. \tag{7.7}$$

The necessary implication of Proposition 2.2 disproves the downstairs estimate. □

Corollary 7.2. *For every $k \geq 1$, a counterexample in range (i) exists with $m = 2, n = k+1, p = n, q = k, a = b = 2$, even on a holomorphically trivial family.*

Proof. Append factors $Z_j = \mathbb{P}^1$, $1 \leq j \leq k$, with fixed Fubini–Study metrics and flat trivial coefficient lines. Write Z_0 for the original factor, whose adjoint coefficient is trivial. In the Künneth decomposition only the multi-index $(0,1,\dots,1)$ contributes, since $H^0(\mathbb{P}^1, K) = 0$ and $H^1(\mathbb{P}^1, K) = \mathbb{C}$. Thus

$$H^k\left(\prod_{j=0}^{k} Z_j, K_{\prod Z_j} \otimes L_y\right) \simeq H^0(Z_0, K_{Z_0} \otimes L_y) \otimes \bigotimes_{j=1}^{k} H^1(Z_j, K_{Z_j}) \simeq \mathbb{C}.$$

Let $\eta_j = \omega_{Z_j}/\operatorname{Vol}(Z_j)$. These are harmonic representatives of trace one, with $\|\eta_j\|^2 = \operatorname{Vol}(Z_j)^{-1}$. The fiber Laplacian on the product is the sum of the factor Laplacians; hence the wedge of the original holomorphic generator with $\eta_1, \dots, \eta_k$ is harmonic. Product integration gives

$$H_k(t,s) = H(t,s) \prod_{j=1}^{k} \operatorname{Vol}(Z_j)^{-1}, \qquad i\Theta_{H_k} = i\Theta_H.$$

The curvature eigenvalues upstairs are $\lambda_-, \lambda_+, 6, 0, \dots, 0$. In dimension $N = k+3$, each $(N, N-1)$ curvature eigenvalue is a sum omitting one of these numbers. Omitting λ_-, λ_+, 6, or one of the zeros gives, respectively,

$$6+\lambda_+, \qquad 6+\lambda_-, \qquad c, \qquad 6+c,$$

all positive. The upstairs estimate follows from Proposition 2.2, whereas the downstairs curvature trace remains -6 at the origin. □

The metric (7.3) is not semipositive: its first curvature block has negative determinant wherever $B \neq 0$.

Proposition 7.3. *There is a proper holomorphic submersion between complete Kähler manifolds admitting positive auxiliary lines for which*

$$(\mathcal{O}_X, 1) \text{ is optimal } (1,2), \qquad (R^1\pi_*\mathcal{O}_X, H) \text{ is not optimal } (1,1).$$

This is range (ii) with $m = n = a = b = q = 1, p = 0$, and the coefficient metric is flat.

Proof. Let

$$Y = \{|t - t_0| < R\} \Subset \{\operatorname{Im} t > 0\}, \quad \psi_Y = -R^2 \log(1 - |t-t_0|^2/R^2), \quad \omega_Y = i\partial\bar{\partial}\psi_Y.$$

Form the holomorphic quotient

$$X = (Y \times \mathbb{C})/\mathbb{Z}^2, \qquad (a,b)\cdot(t,z) = (t, z + a + bt).$$

The action is free: $a+bt=0$ and $\operatorname{Im} t>0$ imply $b=a=0$. It is properly discontinuous. Indeed, on any compact subset of $Y\times\mathbb{C}$, $\operatorname{Im} t\ge c>0$ and $|a+bt|\le C$ force

$$|b|\le C/c,\qquad |a|\le C+|b|\sup|t|,$$

so only finitely many lattice translates can meet that compact set. The quotient is therefore a complex manifold, and the quotient projection to Y is a holomorphic submersion. The real coordinates $z=x+tu$, $x,u\in\mathbb{R}/\mathbb{Z}$, identify X smoothly with $Y\times(\mathbb{R}/\mathbb{Z})^2$, proving that $\pi:X\to Y$ is a proper holomorphic submersion.

Write $v=\operatorname{Im} t$, $w=\operatorname{Im} z$, and $F=w^2/v$. Its complex Hessian in the order (t,z) is

$$(F_{\alpha\bar\beta})=\frac{1}{2v}\begin{pmatrix} w^2/v^2 & -w/v\\ -w/v & 1\end{pmatrix}.$$

It has rank one and is nonnegative. Thus, on the cover,

$$\Omega=i\partial\bar\partial\frac{w^2}{v}=\frac{i}{2v}\left(dz-\frac{w}{v}dt\right)\wedge\overline{\left(dz-\frac{w}{v}dt\right)}. \tag{7.8}$$

Lattice translation changes the potential by $2bw+b^2v=\operatorname{Im}(2bz+b^2t)$, so Ω descends. It is closed and semipositive, with positive vertical restriction; therefore $\omega_X=\Omega+\pi^*\omega_Y$ is Kähler. Moreover,

$$dz-(w/v)dt=dx+t\,du,\qquad \Omega=dx\wedge du,\qquad \int_{X_t}\Omega=1.$$

For the associated distances,

$$\omega_X\ge\pi^*\omega_Y\quad\Longrightarrow\quad d_Y(\pi(x),\pi(x'))\le d_X(x,x').$$

If (x_ν) is d_X-Cauchy, then $\pi(x_\nu)\to y\in Y$. Choose a compact neighborhood K of y containing the projected tail. Properness makes $\pi^{-1}(K)$ compact, so some subsequence converges to x. For any $\varepsilon>0$, choose a sufficiently late subsequence term with $d_X(x_{\nu_j},x)<\varepsilon/2$ and use the Cauchy condition to obtain $d_X(x_\nu,x)\le d_X(x_\nu,x_{\nu_j})+d_X(x_{\nu_j},x)<\varepsilon$. Thus (X,ω_X) is complete.

For the positive auxiliary line, take the quotient with automorphy factor

$$(a,b)\cdot(t,z,\xi)=(t,z+a+bt,e^{-\pi ib^2t-2\pi ibz}\xi).$$

If $j_{a,b}(t,z)=e^{-\pi ib^2t-2\pi ibz}$, then

$$j_{a,b}(t,z+a'+b't)j_{a',b'}(t,z)=e^{-2\pi iba'}j_{a+a',b+b'}(t,z)=j_{a+a',b+b'}(t,z).$$

This is the cocycle identity. The metric descends because

$$|j_{a,b}(t,z)|^2e^{-2\pi(w+bv)^2/v}=e^{2\pi b^2v+4\pi bw}e^{-2\pi(w+bv)^2/v}=e^{-2\pi w^2/v}.$$

Its curvature is $i\partial\bar\partial(2\pi w^2/v)=2\pi\Omega$. Tensoring by the pullback of $e^{-\psi_Y}$ gives positive curvature $2\pi\Omega+\pi^*\omega_Y$. For the coefficient line $(\mathcal{O}_X,1)$, the curvature commutator is zero, so its upstairs estimate follows from Lemma 2.3 after auxiliary twisting.

Under a lattice translation, dz changes to $dz+b\,dt$. Its class modulo base differentials is therefore invariant, defining a holomorphic relative form. On every fiber it spans $H^0(X_t,K_{X_t})$, so it frames $\pi_*K_{X/Y}$, and

$$\omega_t=\frac{i}{2v}dz\wedge d\bar z,\quad \int_{X_t}\omega_t=1,\quad \|dz\|^2=2v.$$

Relative Serre duality gives the holomorphic identification

$$R^1\pi_*\mathcal{O}_X\simeq(\pi_*K_{X/Y})^*.$$

For the pairing $\int\alpha\wedge dz$, the holomorphic dual frame has harmonic representative

$$\alpha_t=\frac{d\bar z}{2iv},\qquad \int_{X_t}d\bar z\wedge dz=2iv,\qquad H(t)=\|\alpha_t\|^2=\frac{1}{2v}.$$

The constant-coefficient form α_t is harmonic for the flat fiber metric. The equation $\int_{X_t} \alpha_t \wedge dz = 1$ shows that its class is the holomorphic dual frame, even though its representative depends on $\operatorname{Im} t$. Explicitly, $|d\bar{z}|^2 = 2v$ and the fiber area is one, so

$$\|\alpha_t\|^2 = (4v^2)^{-1}(2v) \int_{X_t} \omega_t = (2v)^{-1}.$$

Finally $\partial_t v = -i/2$, $\bar{\partial}_t v = i/2$, and $\partial_t \bar{\partial}_t \log v = -1/(4v^2)$. Consequently

$$i\Theta_H = i\partial\bar{\partial} \log(2v) = -\frac{i\, dt \wedge d\bar{t}}{4v^2} < 0.$$

Proposition 2.2 proves the claimed failure. □

8. Scalar-factor metrics and minimal extension

The singular product metric and the metric in the top-adjoint rigidity theorem have the common form

$$E_0 = G \otimes_{\mathbb{C}} V, \qquad H = h_G \otimes H_V, \qquad \mathcal{F} = (G \otimes \mathcal{I}(h_G)) \otimes_{\mathbb{C}} V, \tag{8.1}$$

where (V, H_V) is a fixed finite-dimensional Hermitian space and h_G has psh weights. Pointwise values of H are defined using the psh representatives of these weights.

For a torsion-free sheaf with a singular metric, Griffiths semipositivity means that $\log |\ell|^2_{H^*}$ has a psh representative for every local holomorphic section $\ell \in \mathcal{F}^* = \mathcal{H}om(\mathcal{F}, \mathcal{O}_Y)$. The *minimal extension property* means that there is a proper analytic $S \subset Y$ such that $\mathcal{F}$ is locally free off S and, for every open immersion $\iota : B \hookrightarrow Y$ of the unit ball with $\iota(0) \notin S$ and every $v \in \mathcal{F} \otimes \mathbb{C}(\iota(0))$ with $|v|_H = 1$, there is $\sigma \in H^0(B, \iota^*\mathcal{F})$ with

$$\sigma(0) = v, \qquad \frac{1}{\operatorname{Vol}(B)} \int_B |\sigma|^2_{\iota^* H}\, d\lambda \le 1.$$

The measure here is Euclidean; see [Tak24, Definition 1.2].

Proposition 8.1. *The metric (8.1) is Griffiths semipositive and has the minimal extension property. Moreover, $\mathcal{F} = \mathscr{E}(H)$, and every holomorphic section on the complement of a nowhere-dense analytic set with finite H-norm on every relatively compact ambient neighborhood extends uniquely into $\mathcal{F}$.*

Proof. Integrable sections and removal across analytic sets.

Fix an H_V-orthonormal basis $v_1, \dots, v_d$ and write $s = \sum_\nu f_\nu e_G \otimes v_\nu$ in a local frame with $|e_G|^2_{h_G} = e^{-\varphi}$. Then

$$|s|^2_H = e^{-\varphi} \sum_{\nu=1}^{d} |f_\nu|^2, \qquad |s|^2_H \in L^1_{\mathrm{loc}} \iff f_\nu \in \mathcal{I}(\varphi) \quad (1 \le \nu \le d).$$

This proves $\mathscr{E}(H) = \mathcal{F}$ coefficientwise. If s is given off a nowhere-dense analytic subset A and has finite weighted norm across A, the local upper bound $\varphi \le C$ gives

$$\int_{K \setminus A} \sum_\nu |f_\nu|^2 \le e^C \int_{K \setminus A} |s|^2_H < \infty$$

for compact coordinate neighborhoods K. Extend the coefficients measurably over A, a set of measure zero. Lemma 2.5 with $g = 0$ gives $\bar{\partial} f_\nu = 0$ distributionally on the whole neighborhood, hence f_ν has a holomorphic representative. For the resulting extension $\widetilde{s}$,

$$\int_K |\widetilde{s}|^2_H = \int_{K \setminus A} |s|^2_H < \infty, \qquad \widetilde{s} \in \mathscr{E}(H) = \mathcal{F}.$$

Holomorphic uniqueness on the dense complement makes these local extensions glue.

The dual metric on the actual sheaf. To compute the sheaf dual correctly, write locally $h_G = e^{-\varphi}$ and remove the divisorial part of $\mathcal{I}(\varphi)$:

$$\mathcal{I}(\varphi) = g_D \mathcal{J}, \qquad \psi|_{U\setminus D} = \varphi - \log|g_D|^2, \qquad \operatorname{codim} \operatorname{Supp}(\mathcal{O}/\mathcal{J}) \geq 2.$$

Here ψ denotes the psh extension across D. These are the divisorial-removal identities in the proof of Theorem 1.3, applied to the local psh weight on the base. Put $A = \operatorname{Supp}(\mathcal{O}/\mathcal{J})$. A morphism $\mathcal{J} \to \mathcal{O}$ restricts on $U \setminus A$ to multiplication by a holomorphic function. Hartogs extension across $\operatorname{codim} A \geq 2$ extends that function to U; equality on $U \setminus A$ then determines the morphism everywhere. Therefore

$$\mathcal{H}om(\mathcal{J}, \mathcal{O}) = \mathcal{O}, \qquad \mathcal{F}^{**} = G(-D) \otimes V, \qquad \mathcal{F}^* = G^{-1}(D) \otimes V^*.$$

In the local frame $g_D e_G$ of $G(-D)$, the metric is $e^{-\psi} H_V$, since $|g_D|^2 e^{-\varphi} = e^{-\psi}$. Consequently, for a dual section $\ell = (\ell_\nu)$,

$$\log|\ell|^2_{H^*} = \psi + \log \sum_\nu |\ell_\nu|^2,$$

which is psh. More explicitly, on the set where $F = \sum |\ell_\nu|^2 > 0$, for a tangent vector ξ one has

$$(\partial\bar\partial \log F)(\xi, \bar\xi) = \frac{F \sum_\nu |\partial_\xi \ell_\nu|^2 - |\sum_\nu \bar\ell_\nu \partial_\xi \ell_\nu|^2}{F^2} \geq 0.$$

The holomorphic logarithmic norm has its psh extension over $F = 0$, and adding the psh weight ψ proves Griffiths semipositivity on $\mathcal{F}$.

Extension with the sharp constant. Take $S = \operatorname{Supp}(\mathcal{O}_Y/\mathcal{I}(h_G))$. This is a proper analytic set, and $\mathcal{F} = E_0$ on $Y \setminus S$. On an embedded ball B, the line bundle $\iota^* G$ is trivial: the exponential sequence, Cartan's theorem B and contractibility give

$$H^1(B, \mathcal{O}_B^*) \simeq H^2(B, \mathbb{Z}) = 0.$$

Choose a frame with psh weight φ. Write a unit vector at the center as $v = \sum c_\nu e_G(0) \otimes v_\nu$. Then

$$\varphi(0) > -\infty, \qquad e^{-\varphi(0)} \sum_\nu |c_\nu|^2 = 1.$$

Sharp scalar extension [GZ15] gives a holomorphic F on B such that

$$F(0) = 1, \qquad \int_B |F|^2 e^{-\varphi}\, d\lambda \leq \operatorname{Vol}(B) e^{-\varphi(0)}.$$

Define the holomorphic section $\sigma = \sum c_\nu F e_G \otimes v_\nu$. It satisfies

$$\sigma(0) = v, \qquad \int_B |\sigma|^2_H = \left(\sum_\nu |c_\nu|^2\right) \int_B |F|^2 e^{-\varphi} \leq \operatorname{Vol}(B).$$

The same integral bound gives

$$c_\nu F \in \mathcal{I}(\varphi) \quad (1 \leq \nu \leq d), \qquad \sigma \in H^0(B, \iota^* \mathcal{F}).$$

□

Example 8.2 (A degreewise estimate does not imply minimal extension)**.** Take $Y = \mathbb{C}^2$ with its Euclidean metric, $Z = \mathbb{P}^1$, and a trivial fiber coefficient. On the trivial base line put

$$h_G = e^{-\varphi}, \qquad \varphi(y) = -|y_1|^2 + 2|y_2|^2.$$

For $(m, n, a, b, p, q) = (2, 1, 2, 2, 1, 1)$ the upstairs curvature eigenvalues are $-1, 2, 0$, so $C_{L;3,3} = 1$. The direct image is $G \otimes H^1(\mathbb{P}^1, K_{\mathbb{P}^1}) \simeq G$, with metric h_G up to a constant; it satisfies the optimal $(2, 2)$ estimate, but has a negative curvature direction.

Minimal extension fails at every center, which is necessary to exclude an exceptional analytic set. Given y^0, pull back to the unit ball by $\iota(w) = (y_1^0 + \varepsilon w_1, y_2^0 + \delta w_2)$, where $\varepsilon^2 > 2\delta^2 > 0$. The pulled-back weight is

$$\varphi \circ \iota = \varphi(y^0) + 2\,\mathrm{Re}\,\ell(w) - \varepsilon^2|w_1|^2 + 2\delta^2|w_2|^2, \qquad \ell(w) = -\overline{y_1^0}\varepsilon w_1 + 2\overline{y_2^0}\delta w_2.$$

In the holomorphic frame $e' = e^{\varphi(y^0)/2+\ell(w)}e_G$, the squared norm is

$$W(w) = \exp(\varepsilon^2|w_1|^2 - 2\delta^2|w_2|^2), \qquad W(0) = 1.$$

Any extension of the unit vector $e'(0)$ is Fe' with $F \in \mathcal{O}(B)$ and $F(0) = 1$. Write $F = \sum_{\alpha\in\mathbb{N}^2} c_\alpha w^\alpha$, so $c_0 = 1$. For $B_r = \{|w| < r\}$, $0 < r < 1$, and $\alpha \neq \beta$, independent coordinate rotations give

$$\int_{B_r} w^\alpha \bar{w}^\beta W\, d\lambda = e^{i(\alpha-\beta)\cdot\theta} \int_{B_r} w^\alpha \bar{w}^\beta W\, d\lambda = 0$$

by choosing θ with $e^{i(\alpha-\beta)\cdot\theta} \neq 1$. Uniform convergence of the Taylor series on $\overline{B_r}$ gives

$$\int_{B_r} |F|^2 W\, d\lambda = \sum_\alpha |c_\alpha|^2 \int_{B_r} |w^\alpha|^2 W\, d\lambda \geq \int_{B_r} W\, d\lambda.$$

Letting $r \uparrow 1$ preserves the inequality by monotone convergence. Rotational symmetry and polar integration on the unit ball give $\mathrm{Vol}(B)^{-1} \int_B |w_j|^2\, d\lambda = 1/3$. Jensen's inequality now yields

$$\frac{1}{\mathrm{Vol}(B)} \int_B |F|^2 W \geq \frac{1}{\mathrm{Vol}(B)} \int_B W \geq \exp\left(\frac{\varepsilon^2 - 2\delta^2}{3}\right) > 1.$$

This contradicts the minimal bound at the arbitrary center y^0. Hence the set of failure is all of Y and cannot be contained in the exceptional analytic set allowed in the definition.

References

[BP08] B. Berndtsson and M. Păun, *Bergman kernels and the pseudoeffectivity of relative canonical bundles*, Duke Math. J. **145** (2008), no. 2, 341–378. https://doi.org/10.1215/00127094-2008-054.

[BPW22] Bo Berndtsson, Mihai Păun, and Xu Wang, *Algebraic fiber spaces and curvature of higher direct images*, J. Inst. Math. Jussieu **21** (2022), no. 3, 973–1028. doi:10.1017/S147474802000050X.

[Ber09] B. Berndtsson, *Curvature of vector bundles associated to holomorphic fibrations*, Ann. of Math. (2) **169** (2009), no. 2, 531–560. doi:10.4007/annals.2009.169.531.

[CP17] J. Cao and M. Păun, *Kodaira dimension of algebraic fiber spaces over abelian varieties*, Invent. Math. **207** (2017), no. 1, 345–387. https://doi.org/10.1007/s00222-016-0672-6.

[CS24] Y.-J. Choi and G. Schumacher, *Curvature of higher direct images of sheaves of twisted holomorphic forms*, preprint (2024), arXiv:2407.04298.

[DEL00] J.-P. Demailly, L. Ein, and R. Lazarsfeld, *A subadditivity property of multiplier ideals*, Michigan Math. J. **48** (2000), 137–156. arXiv:math/0002035v2; see Theorem 2.6(i) for the analytic product formula.

[DNWZ23] Fusheng Deng, Jiafu Ning, Zhiwei Wang, and Xiangyu Zhou, *Positivity of holomorphic vector bundles in terms of L^p-estimates for $\bar{\partial}$*, Math. Ann. **385** (2023), nos. 1–2, 575–607. doi:10.1007/s00208-021-02348-7.

[DPS01] J.-P. Demailly, T. Peternell, and M. Schneider, *Pseudo-effective line bundles on compact Kähler manifolds*, Internat. J. Math. **12** (2001), no. 6, 689–741. https://doi.org/10.1142/S0129167X01000861.

[Dem12] J.-P. Demailly, *Analytic Methods in Algebraic Geometry*, Surveys of Modern Mathematics, vol. 1, International Press and Higher Education Press, 2012.

[Dem82] J.-P. Demailly, *Estimations L^2 pour l'opérateur $\bar{\partial}$ d'un fibré vectoriel holomorphe semi-positif au-dessus d'une variété kählérienne complète*, Ann. Sci. École Norm. Sup. (4) **15** (1982), no. 3, 457–511. Full text; see §5, Theorem 5.1.

[Fuj78] T. Fujita, *On Kähler fiber spaces over curves*, J. Math. Soc. Japan **30** (1978), no. 4, 779–794. doi:10.2969/jmsj/03040779.

[GZ15] Q. Guan and X. Zhou, *A solution of an L^2 extension problem with an optimal estimate and applications*, Ann. of Math. (2) **181** (2015), no. 3, 1139–1208. doi:10.4007/annals.2015.181.3.6.

[Gra60] Hans Grauert, *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Inst. Hautes Études Sci. Publ. Math. **5** (1960), 5–64. doi:10.1007/BF02684746.

[HPS18] C. Hacon, M. Popa, and C. Schnell, *Algebraic fiber spaces over abelian varieties: around a recent theorem by Cao and Păun*, in *Local and Global Methods in Algebraic Geometry*, Contemp. Math. **712**, Amer. Math. Soc., Providence, RI, 2018, 143–195. https://doi.org/10.1090/conm/712/14346.

[IMW24] T. Inayama, S.-i. Matsumura, and Y. Watanabe, *Singular Nakano positivity of direct image sheaves of adjoint bundles*, Anal. PDE, to appear; https://arxiv.org/abs/2407.11412, version 2 (2024).

[Ina26] T. Inayama, *Partial positivity is not preserved by marginalization*, preprint (2026), https://arxiv.org/abs/2609.15075.

[Kaw81] Y. Kawamata, *Characterization of abelian varieties*, Compositio Math. **43** (1981), no. 2, 253–276.

[Kol86a] J. Kollár, *Higher direct images of dualizing sheaves. I*, Ann. of Math. (2) **123** (1986), no. 1, 11–42. doi:10.2307/1971351.

[Kol86b] J. Kollár, *Higher direct images of dualizing sheaves. II*, Ann. of Math. (2) **124** (1986), no. 1, 171–202.

[LY14] Kefeng Liu and Xiaokui Yang, *Curvatures of direct image sheaves of vector bundles and applications*, J. Differential Geom. **98** (2014), no. 1, 117–145. arXiv:1303.2701.

[MT07] C. Mourougane and S. Takayama, *Hodge metrics and positivity of direct images*, J. Reine Angew. Math. **606** (2007), 167–178. https://doi.org/10.1515/CRELLE.2007.039.

[MT08] Ch. Mourougane and S. Takayama, *Hodge metrics and the curvature of higher direct images*, Ann. Sci. École Norm. Sup. (4) **41** (2008), no. 6, 905–924. arXiv:0707.3551v2; doi:10.24033/asens.2084.

[MT09] C. Mourougane and S. Takayama, *Extension of twisted Hodge metrics for Kähler morphisms*, J. Differential Geom. **83** (2009), no. 1, 131–161. https://arxiv.org/abs/0809.3221.

[Mat22] S.-i. Matsumura, *Injectivity theorems with multiplier ideal sheaves for higher direct images under Kähler morphisms*, Algebr. Geom. **9** (2022), no. 2, 122–158. https://doi.org/10.14231/AG-2022-005.

[Nau21] Philipp Naumann, *Curvature of higher direct images*, Ann. Fac. Sci. Toulouse Math. (6) **30** (2021), no. 1, 171–201. doi:10.5802/afst.1670.

[PT18] M. Păun and S. Takayama, *Positivity of twisted relative pluricanonical bundles and their direct images*, J. Algebraic Geom. **27** (2018), no. 2, 211–272. https://doi.org/10.1090/jag/702.

[RRV71] J.-P. Ramis, G. Ruget, and J.-L. Verdier, *Dualité relative en géométrie analytique complexe*, Invent. Math. **13** (1971), 261–283. https://doi.org/10.1007/BF01406078.

[Siu74] Y.-T. Siu, *Analyticity of sets associated to Lelong numbers and the extension of closed positive currents*, Invent. Math. **27** (1974), 53–156. https://doi.org/10.1007/BF01389965.

[Tak24] S. Takayama, *Singular Griffiths semi-positivity of higher direct images*, Math. Ann. **388** (2024), 4343–4353. doi:10.1007/s00208-023-02632-8.

[Take95] K. Takegoshi, *Higher direct images of canonical sheaves tensorized with semi-positive vector bundles by proper Kähler morphisms*, Math. Ann. **303** (1995), 389–416.

[Vie83] E. Viehweg, *Weak positivity and the additivity of the Kodaira dimension for certain fibre spaces*, in *Algebraic varieties and analytic varieties (Tokyo, 1981)*, Adv. Stud. Pure Math., vol. 1, North-Holland, Amsterdam, 1983, pp. 329–353. doi:10.2969/aspm/00110329.

[Wat25a] Y. Watanabe, *Dual Nakano positivity and singular Nakano positivity of direct image sheaves*, Nagoya Math. J. **257** (2025), 55–78. doi:10.1017/nmj.2024.20.

[Wat25b] Yuta Watanabe, *Curvature operator of holomorphic vector bundles and L^2-estimate condition for (n,q) or (p,n)-forms*, Tohoku Math. J. (2) **77** (2025), no. 4, 445–471. doi:10.2748/tmj.20231128. Theorem numbers cited here refer to arXiv:2109.12554v2.

[Wat25c] Y. Watanabe, *Nakano–Nadel type, Bogomolov–Sommese type vanishing and singular dual Nakano semi-positivity*, Ann. Fac. Sci. Toulouse Math. (6) **34** (2025), no. 2, 339–394. https://doi.org/10.5802/afst.1815; arXiv:2209.00823v2. The weighted Dolbeault resolution is Theorem 5.3(a) in the cited arXiv version.

[Wu21] X. Wu, *On the hard Lefschetz theorem for pseudoeffective line bundles*, Internat. J. Math. **32** (2021), no. 6, Paper No. 2150035, 25 pp. https://doi.org/10.1142/S0129167X2150035X.

Institute of Mathematics, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: chenzhijian@amss.ac.cn

Beijing Institute of Mathematical Sciences and Applications; Department of Mathematics and Yau Mathematical Sciences Center, Tsinghua University, Beijing, China

Email address: liuzhuo@amss.ac.cn

Yau Mathematical Sciences Center, Tsinghua University, Beijing 100084, China

Email address: qingchenghao@amss.ac.cn